Preliminary: Please do not quote or cite

Quota Baskets Versus Tax Baskets in Fisheries

Click here for the latest version of the paper

Mauricio Collado *

November 9, 2026

Abstract

Because managing stocks individually is costly, many fisheries bundle multiple stocks and manage them with a single instrument. I study when such bundling is desirable and whether it should take the form of a shared quota or a shared landing fee, extending the work of Reed (1979) and Weitzman (2002) to a multispecies fishery under ecological uncertainty.

A well-designed basket groups species whose marginal profits at their own optimal escapements are similar, a criterion that combines biological productivity, harvesting costs, and prices. Under ecological uncertainty, the tax basket outperforms the quota basket. However, as the number of species in the group increases and shocks are independent across species, the quota's uncertainty loss is diversified away per species, and the total welfare gap settles at a positive constant. Yet increasing the basket size has a second cost: the mismatch from pooling heterogeneous species grows with the number of species for both instruments. Larger baskets are not always better.

Bundling improves on species-specific management only when the monitoring cost savings exceed this mismatch loss.

JEL Classification: Q22, Q28, Q58, D81

1 Introduction

Allocating property rights through species-specific quotas has been shown to mitigate the common-pool resource problem in fisheries (Costello et al., 2016; FAO, 2025; Hilborn et al., 2020; Hoshino et al., 2020; Mangin et al., 2018). Without quotas, competing fishermen have an incentive to harvest more today, depleting the biomass available tomorrow (Clark, 1990;

*University of California, Santa Barbara (nestorcollado@bren.ucsb.edu). This work was made possible by the support and suggestions of Chris Costello, Andrew Plantinga, Antony Millner, Costello's Lab, and my colleagues at Econ 290.

Gordon, 1954). Despite the success of catch shares, typically only wealthy countries can bear the costs of fully implementing them for many species (Mangin et al., 2018). Because most commercial fisheries harvest multiple species jointly (Fulton et al., 2022; Spence et al., 2021), some countries instead pool many stocks and regulate them as one (Kulanujaree et al., 2020) or apply a single economic instrument to the whole group, a quota basket (Sanchirico et al., 2006). Quota baskets have historically been dismissed because they risk overfishing vulnerable species (Bonzon et al., 2013; Clark and Kirkwood, 1986), yet there is evidence that they can be economically and ecologically viable (Collado et al., 2021; Karr et al., 2021; Kritzer et al., 2023; Kulanujaree et al., 2020). Bundling trades a blunter instrument for lower administrative costs. How do we reconcile the overfishing risk with the viability observed in practice?

In this paper, I ask when it is desirable to bundle species under a single instrument and whether that instrument should be a shared quota or a uniform landing fee. To compare the two, I introduce the tax basket, a single landing fee applied to a group of species. To my knowledge, no fishery relies exclusively on landing fees, but the tax basket isolates how each instrument handles ecological uncertainty, because the optimal landing fee does not depend on ecological shocks (Weitzman, 2002). I extend Reed's (1979) constant-escapement model and Weitzman's (2002) information framework to a fishery with N heterogeneous, ecologically independent stocks, one fisherman, one social planner, and one basket. The planner sets the instrument before ecological shocks are realized; the fisherman harvests after observing them. Under either instrument, the fisherman equalizes marginal profits across species, so designing a basket amounts to choosing one common margin for all species. I benchmark both baskets against species-specific landing fees, which implement Reed's escapement for every stock, and decompose the welfare loss into a mismatch loss (from imposing one margin on heterogeneous species) and an uncertainty loss (from letting the margin move with realized shocks). To isolate the instrument's role, the model abstracts from management costs, gear choice, ecological interactions, strategic behavior, and bycatch.

I obtain three main results. First, the welfare-maximizing basket margin is a weighted average of the species' next-period marginal profits, the fisheries analog of the optimal uniform tax across heterogeneous pollutants. As an intertemporal equimarginal principle, the margin combines biological productivity, harvesting costs, and prices. Second, Weitzman's ranking survives bundling: the tax basket yields higher expected welfare than the quota basket, because the tax fixes the common margin directly, while the quota's shadow value moves with the aggregate shock. Third, basket size has two effects. Independent shocks partly cancel in the aggregate, so the quota basket's disadvantage per species falls at rate $1/N$, although its total disadvantage does not vanish. When a common environmental driver correlates the shocks, diversification is only partial, and the per-species disadvantage converges to a positive floor. At the same time, the mismatch loss grows roughly linearly with N for both instruments. Simulations at the typical recruitment variability of exploited fish stocks (Thorson et al., 2014) illustrate this trade-off: the uncertainty loss is the larger

of the two in a two-species basket, but the mismatch loss is already twice as large at five species and roughly thirty times as large at fifty. Larger baskets are not always better: bundling pays only when the monitoring savings exceed these efficiency losses.

The case for bundling rests on three difficulties of species-specific regulation in multispecies fisheries, which stem from joint production, limited selectivity, imperfect information, and fishing gear choice (Branch, 2009; Christensen and Pauly, 2004; Copes, 1986; May and Clark, 1979; Nakamura, 2015; Pauly et al., 1998; Sanchirico et al., 2006). First, species-specific regulation ignores natural clusters (shoals) and species interactions through ecology, regulation, and non-selective fishing gear. As a consequence, it omits the trade-off between sustainability and profit maximization created by choke species: the quota of the most restrictive species leads to premature season closures (Abbott and Wilen, 2009) and blocks the extraction and revenue of non-vulnerable species (Hatcher, 2022; Ulrich et al., 2017; Wo et al., 2022). This was the case for co-occurring groundfish fisheries on the U.S. West Coast. Hilborn et al. (2004) modeled twelve species in this fishery and found that maximizing profits meant overfishing some species, whereas preventing overfishing of the most vulnerable stocks caused a 90% loss in yield. Mixed fisheries have repeatedly failed to achieve the intertemporal profit maximization that characterizes species-specific catch-share management.

Second, Tinbergen's (1952) rule calls for one corrective instrument per market failure, but planners are constrained by how many economic instruments they can credibly design, monitor, enforce, and govern. Fully implementing regulation for even one species is data-intensive (Mangin et al., 2018; Su et al., 2020), costly to administer (Karr et al., 2021), and politically unstable (OECD, 2003). Even when the resources are available, species are hard to identify at port or during the fishing season (Pacific Fishery Management Council, 2024; Squires et al., 1998) or are aggregated in datasets (Su et al., 2020; Sun et al., 2023). Such problems are especially pronounced in Chinese fisheries, which often harvest a broad mix of species with minimal selectivity, leading to indiscriminate removal across trophic levels (X. Jin et al., 2020; Kritzer et al., 2023; Szuwalski et al., 2017). Low selectivity hampers control and monitoring (Sun et al., 2023; Ye et al., 2013), while aggregation in official landings data complicates identifying stock and harvest composition (Cao et al., 2017; Pauly and Froese, 2012). These issues limit the feasibility of traditional species-specific management that relies on precise stock assessments.

Third, applying species-specific instruments to a multispecies group goes beyond administrative difficulty: the mismatch also causes concrete harm. It induces discarding of low-value species, or high-grading (Copes, 1986; Ulrich et al., 2017), effort reallocation toward unregulated stocks (Smith et al., 2009), and intertemporal arbitrage (Holzer and DePiper, 2019).

This aggregation problem belongs to a broader class of second-best regulation problems in which a single instrument must address multiple market failures. In environmental economics, this question has been studied for uniform versus differentiated taxes across heterogeneous pollution sources in a static setting (Diamond, 1973; Fowlie and Muller,

2019; Mendelsohn, 1986), for the choice between prices, quantities, and mixed policies when multiple pollutants are regulated jointly (Ambec and Coria, 2013), and for single taxes versus single quotas under uncertainty (Hoel and Karp, 2002). The fisheries literature has also proposed alternatives for handling multiple stocks, such as a shared cap set as a percentage of the sum of individual caps (Sanchirico et al., 2006), a shared cap that targets a group mortality (Garcia et al., 2020), a shared cap with safeguard caps for vulnerable species (Sun et al., 2025), multispecies maximum sustainable yield (Fulton et al., 2022), aggregate production models (Fogarty et al., 2012), quota exchanges (Singh and Weninger, 2024), and size-spectrum models (Andersen et al., 2016), among others. Most of these methods help, but they are prohibitive for many regulators because of limited scientific capacity, information, enforceability, and budget. Moreover, many do not incorporate ecological uncertainty.

Unlike these solutions, quota baskets do not rely on a single-species framework or rich datasets (Collado et al., 2024). They bundle species into a single instrument, allowing flexibility in harvest allocation by trading stringency for convenience. However, theoretical and empirical research on this topic remains scarce. Clark (1990) showed that joint harvesting could not sustain both species in the short run, depleting the more vulnerable stock in a static open-access setting. Extending the analysis dynamically, he demonstrated that joint harvesting may be feasible but leads to an unstable equilibrium. In a non-fishery setting, Weitzman (1998) examined biodiversity preservation under budget constraints and showed that optimal bundling strategies may imply sacrificing some species to preserve others. More recently, Collado et al. (2021) analyzed the fisherman's choice of fishing gear under multiple basket allocations in a modified Gordon–Schaefer model. The authors suggest that well-designed baskets can maintain moderate overall stock survival and intertemporal profit maximization, whereas poorly designed configurations risk the collapse of the vulnerable species or of all species in the group.

Consensus also remains lacking on how to group species and on the conditions needed for a welfare-maximizing quota basket. Previous research recommends grouping species by similarity in ecological and economic traits (Collado et al., 2021; Costello and Szuwalski, 2018), by similar vulnerability indicators (catchability over intrinsic growth rate) (Collado et al., 2024), or by relying on local fishermen's expertise (EDF, 2020). Using empirical data from Belizean fisheries (United Nations Conference on Trade and Development, 2022), Collado et al. (2024) found that the optimal quota basket was approximately 30% of the sum of individual maximum sustainable yields for selected baskets. These efforts build intuition but do not provide a definitive answer, especially in more complex settings such as those with ecological uncertainty.

To my knowledge, this is the first paper to compare quota baskets with tax baskets. It extends the seminal work of Reed (1979) and Weitzman (2002) to a multispecies setting and speaks to the literature on regulating multiple pollutants (Ambec and Coria, 2013; Diamond, 1973; Fowlie and Muller, 2019; Mendelsohn, 1986). The optimal basket margin is the fisheries analog of the optimal uniform tax across heterogeneous pollutants, where the

tax is a weighted average of marginal damages. Most of that literature is static; this paper studies a dynamic, stochastic setting. Diversification across species also links basket design to portfolio theory (Edwards et al., 2004; D. Jin et al., 2016; Markowitz, 1952; Sanchirico et al., 2008), but here variance falls by adding species rather than by optimizing portfolio weights.

The paper also contributes to the literature on partial flexibility in multispecies quota systems. Whereas Singh and Weninger (2024) characterize the optimal degree of flexibility given a species-specific quota design, I derive the conditions for full bundling and compare the resulting quota basket with its tax counterpart. The resulting grouping criterion, similarity in individual optimal taxes, offers an alternative to vulnerability indices (Collado et al., 2024) that combine catchability and intrinsic growth.

Finally, the paper contributes to the literature on fisheries under uncertainty (Clark and Kirkwood, 1986; Hannesson and Kennedy, 2003; Reed, 1979; Sethi et al., 2005). The analysis shows that, even under uncertainty, basket instruments can be designed using a well-defined optimality criterion rather than ad hoc grouping rules. I also show when the gap between instruments narrows: with independent shocks, adding species reduces the quota's per-species disadvantage, while the total gap settles at a positive constant. This matters when taxes are not politically viable and the regulator must make quotas as effective as possible.

The remainder of the paper is organized as follows. Section 2 presents the model and theoretical analysis. Section 3 reports the numerical results. Section 4 discusses policy implications and concludes.

2 Model

Throughout the paper, $t = 0, 1, \ldots$ denotes time, and $i = 1, 2, \ldots, N$ indexes the different species. The model relies on two cornerstones: (i) the intertemporal profits for a particular species are maximized with a fixed escapement over time in both deterministic and stochastic settings (Reed, 1979), and (ii) a fixed landing fee induces the fisherman to stop fishing at the optimal escapement (Weitzman, 2002).[1] I model one basket, one fisherman, and one social planner.

For the ecology component, I make three key assumptions. First, I assume the species' life cycles are independent, so each species' growth function depends only on its own remaining stock (escapement) y^i_{t-1} from the previous period. The function follows the usual conventions: it is differentiable, concave $\frac{\partial^2 G^i(y^i_{t-1})}{\partial (y^i_{t-1})^2} \leq 0$, and increasing $\frac{\partial G^i(y^i_{t-1})}{\partial y^i_{t-1}} > 0$. Second, I assume the random ecological shocks $\{Z^i_t\}$ are independent and identically distributed over time, and independent across species, on some interval $[a, b]$,[2] with $\mathbb{E}[Z^i_t] = 1$

[1] An alternative proof can be found in Appendix A.

[2] The population is self-sustaining within the interval $[\underline{x}, \overline{x}]$, where $\overline{x} = \max\{bG(s) : bG(s) \geq s, s \geq 0\}$

and $\mathbb{E}|\ln Z_t^i| < \infty$. Third, I assume the shocks are exogenous to the stocks and the choice variables.[3]

For the fisherman's behavior, I make four key assumptions. First, the fisherman is myopic. This assumption captures the common-pool externality, and the fisherman's behavior is equivalent to a competitive fleet that does not internalize how today's harvest affects tomorrow's stock. Second, I assume a separable harvest technology: the fisherman can choose each species' catch independently, so there is no unwanted catch. In practice, an incidentally caught species in the basket simply counts against the shared cap; this absorption is one reason baskets exist, because it avoids choke species. Bycatch of vulnerable species outside the basket remains a limitation (Section 4). Under this assumption, I do not model the choice of fishing technology or gear. The marginal harvesting cost is positive and decreasing in the available stock s^i, $C^i(s^i) > 0$ and $C^{i\prime}(s^i) < 0$, with time-invariant cost parameters θ^i. I work with the cost function $C^i(s^i) = \frac{\theta^i}{s^i}$, which corresponds to constant catchability. I show that this form lets the tax deliver the planner's target escapements regardless of the realized shocks. Third, prices are strictly positive for all species in the basket; therefore, discards are not considered, as all harvest is valuable. Fourth, all species can be fished in all periods (the fishable set $\mathcal{F}$ defined in Lemma 1). This assumption prevents species from entering or leaving the harvest, which facilitates comparison across instruments.

The social planner is forward-looking and risk-neutral, with a time-invariant discount factor β. For consumers, I assume perfectly elastic demand for all species, so prices are fixed.

I also adopt Weitzman's (2002) information framework, which combines a static optimization problem for a myopic fisherman with a forward-looking social planner:

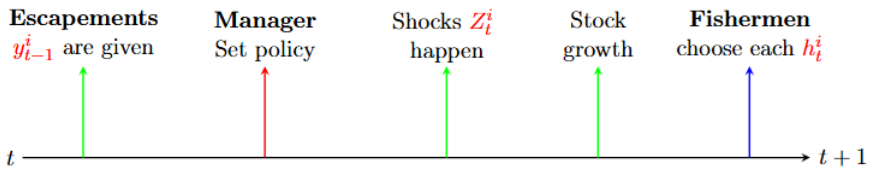


Figure 1: Information flow for each species i between periods t and $t+1$

At each period t, the following sequence of events occurs:

1. Escapements y_{t-1}^i for each species i are predetermined by the choices of the previous period.

2. The social planner sets a basket policy (a quota ω_t or a tax τ_t) without knowing how the stock composition will change with the ecological shocks.

and $\underline{x} = \max\{aG(s) : aG(s) \geq s, s \geq 0\}$.

[3] That is, $\mathbb{E}[Z_t^i \mid \cdot] = 1$ conditional on the present and past values of X_t^i and of the choice variables.

3. The ecological shocks Z_t^i are realized. Each species grows, and the stock shares change.

4. The fisherman observes the realized stocks and chooses the harvest h_t^i, given the basket policy set by the planner.

The key friction is that the planner chooses a basket policy to address multiple common-pool resource problems before shocks are realized, whereas the fisherman makes harvest decisions after shocks are realized. Given the economic instrument, a myopic fisherman observes the realized shocks and chooses harvest to maximize current profit:

1. Under a tax basket:

$$\max_{\{h_t^i\}_{i=1}^N} \sum_{i=1}^{N} \int_{X_t^i - h_t^i}^{X_t^i} (p^i - \tau_t - C^i(s^i))\, ds^i \tag{1}$$

2. Under a quota basket:

$$\begin{aligned} &\max_{\{h_t^i\}_{i=1}^N} \sum_{i=1}^{N} \int_{X_t^i - h_t^i}^{X_t^i} (p^i - C^i(s^i))\, ds^i \\ &\text{subject to} \quad \omega_t = \sum_{i=1}^{N} h_t^i \end{aligned} \tag{2}$$

where X_t^i is the stock at the start of period t, and h_t^i is the harvest. Because the escapement y^i is the reference for species-specific profit maximization, I re-express the integral bounds using the definition of escapement, $y_t^i = X_t^i - h_t^i$:

$$\sum_{i=1}^{N} \int_{X_t^i - h_t^i}^{X_t^i} \left(p^i - C^i(s^i)\right) ds^i = \sum_{i=1}^{N} \int_{y_t^i}^{X_t^i} \left(p^i - C^i(s^i)\right) ds^i$$

The forward-looking social planner, in turn, sets the instrument (ω_t or τ_t) before shocks are realized to maximize expected discounted profits, taking the fisherman's decision rule for each species i as given:

1. Under a tax basket:

$$\begin{aligned} &\max_{\{\tau_t\}} \mathbb{E}\left[\sum_{t=0}^{\infty} \beta^t \sum_{i=1}^{N} \left(\underbrace{\int_{X_t^i - h_t^i(\tau_t)}^{X_t^i} (p^i - \tau_t - C^i(s^i))\, ds^i}_{\text{Species } i \text{ profit}} + \underbrace{\int_{X_t^i - h_t^i(\tau_t)}^{X_t^i} \tau_t\, ds^i}_{\text{Species } i \text{ tax revenue}} \right) \right] \\ &\text{subject to} \quad X_t^i = Z_t^i G^i(y_{t-1}^i) \end{aligned} \tag{3}$$

The tax terms in the profit and revenue components cancel.

2. Under a quota basket:

$$\begin{aligned} &\max_{\{\omega_t\}} \mathbb{E}\left[\sum_{t=0}^{\infty}\beta^t \sum_{i=1}^{N}\int_{X_t^i-h_t^i(\omega_t)}^{X_t^i}\left(p^i - C^i(s^i)\right)ds^i\right] \\ &\text{subject to} \quad X_t^i = Z_t^i G^i(y_{t-1}^i), \quad \omega_t = \sum_{i=1}^{N} h_t^i \end{aligned} \tag{4}$$

where $\beta \in (0,1)$ is the discount factor.

With this procedure, I obtain (1) the fisherman's static decision rule for y_t^i, (2) the planner's dynamic optimal policy choice after incorporating (1), and (3) the fisherman's escapement response to the choice in (2). I can then calculate the welfare difference between instruments. To complete the comparison, I use the species-specific first-best taxes τ^{i*}, which Weitzman (2002) shows do not depend on ecological shocks. I denote by $\mathcal{W}^{\text{ind}}$ the intertemporal sum of profits and tax revenues in this case. The tax basket applies a uniform landing fee τ^+ to the group and yields welfare $\mathcal{W}^{\tau}$; the quota basket applies a single aggregate harvest limit ω_t and yields welfare $\mathcal{W}^{\omega}$. Because the first-best vector of taxes avoids both the uncertainty loss and the mismatch loss (the efficiency loss from forcing a common instrument level on all species), and because I will show that the tax basket avoids the uncertainty loss, the three schemes are ordered $\mathcal{W}^{\text{ind}} \geq \mathcal{W}^{\tau} \geq \mathcal{W}^{\omega}$. Because the planner is risk-neutral, I compare instruments by expected welfare and omit the expectation operator when no confusion arises. I decompose the regulatory loss as

$$\underbrace{\mathcal{W}^{\text{ind}} - \mathcal{W}^{\omega}}_{\text{Regulatory loss}} = \underbrace{\mathcal{W}^{\text{ind}} - \mathcal{W}^{\tau}}_{\text{Mismatch loss}} + \underbrace{\mathcal{W}^{\tau} - \mathcal{W}^{\omega}}_{\text{Uncertainty loss}} \geq 0. \tag{5}$$

The model assumes nothing about the social planner's knowledge of the stock composition at the start of period t. This is deliberate: Section 2.3 shows that the planner does not need data on the initial composition of the biomass.

2.1 Fisherman's decision rule

From the FOC of Equation (1) with respect to each harvest, and using the definition of escapement $y_t^i = X_t^i - h_t^i$, I obtain the fisherman's optimality condition for the current period:

$$p^1 - C^1\left(y_t^1\right) = p^2 - C^2\left(y_t^2\right) = \cdots = p^N - C^N\left(y_t^N\right) = \tau_t \tag{6}$$

Equation (6) is the equimarginal principle: total profits are maximized when the marginal profit of the last harvested unit is equal across species. The fisherman's escapement decision is:

$$
\begin{aligned}
p^i - \tau_t - C^i(y_t^i) &= 0 \\
C^i(y_t^i) &= p^i - \tau_t \\
y_t^i &= (C^i)^{-1}(p^i - \tau_t)
\end{aligned} \tag{7}
$$

In this static problem, the escapement rule also serves as the harvest rule for each species. Given the parameter assumptions, the tax does not depend on the state variables (I verify this in the planner's problem).

In Equation (2), the fisherman faces the same maximization problem but for a given quota basket ω_t. The optimality condition in this case is:

$$
p^1 - C^1\left(y_t^1\right) = p^2 - C^2\left(y_t^2\right) = \cdots = p^N - C^N\left(y_t^N\right) = \lambda_t \tag{8}
$$

This yields the same escapement rule, $y_t^i = (C^i)^{-1}(p^i - \lambda_t) = Y^i(\lambda_t)$. Note, however, that the shadow value is a function of the stock vector $\vec{X}_t$ and the quota basket ω_t, because

$$
\begin{aligned}
\omega_t &= \sum_i X_t^i - \sum_i Y^i(\lambda_t) \\
\sum_i Y^i(\lambda_t) &= \sum_i X_t^i - \omega_t \\
\lambda_t &= \lambda(\vec{X}_t, \omega_t)
\end{aligned} \tag{9}
$$

The fisherman allocates harvest by equalizing marginal profits across all species in the basket at a common margin $m\pi_t$, which equals τ_t under the tax and $\lambda_t(\vec{X}_t, \omega_t)$ under the quota. The fisherman's decision under a quota basket is therefore:

$$
y_t^i = (C^i)^{-1}(p^i - \lambda(\vec{X}_t, \omega_t)) \tag{10}
$$

In this case, the shadow value depends on the state of all stocks, which makes the quota basket more complex than the tax basket. In both cases, the variable that drives the fisherman's decision is the marginal profit, and the tax and the quota are vehicles to set it:

Lemma 1 (Fisherman's optimality condition)**.** *The optimality condition for harvesting N species in the fishable set $\mathcal{F} = \{i : p^i - m\pi > C^i(K^i)\}$, where K^i is the carrying capacity of species i, is:*

$$
p^1 - C^1\left(y_t^1\right) = p^2 - C^2\left(y_t^2\right) = \cdots = p^N - C^N\left(y_t^N\right) = m\pi_t
$$

The stock-dependent cost function incorporates the economic and biological parameters of all species into the optimality condition. The set $\mathcal{F}$ is defined at carrying capacity; whether a species is actually harvested in period t also requires $X_t^i > Y^i(m\pi_t)$. The theory assumes this holds every period. Section 3 reports how often it fails under the quota.

2.2 Optimal marginal profits

The social planner incorporates the fisherman's escapement decision into her dynamic optimization problem:

$$V(\vec{y}_{t-1}) = \max_{m\pi_t \geq 0} \mathbb{E}\left\{ \sum_{i=1}^{N} \int_{Y^i(m\pi_t)}^{Z_t^i G^i(y_{t-1}^i)} \left(p^i - C^i(s^i)\right) ds^i + \beta V(\vec{y}_t) \right\} \tag{11}$$
$$\text{s.t. } X_t^i = Z_t^i G^i(y_{t-1}^i), \quad y_t^i = Y^i(m\pi_t)$$

Taking the FOC with respect to $m\pi_t$ gives the optimal marginal profit:

Proposition 1 (Welfare-maximizing $m\pi$)**.** *Consider N heterogeneous, independent stocks with cost functions C^i, i.i.d. ecological shocks Z_t^i with $\mathbb{E}[Z_t^i] = 1$, and growth functions $G^i(y_{t-1}^i)$. Assume the interior regularity condition*

$$\max_i \tau^{i*} \; < \; \tau_{\max}^F \; := \; \min_i \Big(p^i - C^i(K^i)\Big), \quad K^i \text{ is the carrying capacity,} \tag{12}$$

so that all species remain in the fishable set on $[\min_i \tau^{i*}, \max_i \tau^{i*}]$. Then the optimal $m\pi_t^+$ solves the implicit equation

$$m\pi_t^+ = \mathbb{E}\Big[\sum_{i=1}^{N} \underbrace{\rho_t^i(m\pi_t^+)}_{\text{Weight } i} \quad \underbrace{\pi_{t+1}^i}_{\text{Marginal profit } i \text{ at } t+1}\Big] \tag{13}$$

where $\rho_t^i = \frac{y^i_{m\pi_t^+}}{\sum_{j=1}^N y^j_{m\pi_t^+}}$, with $y_{m\pi}^i := dY^i/dm\pi$ the response of species i's escapement to the margin, and $\pi_{t+1}^i = \beta\big(p^i - C^i(Z_{t+1}^i G^i(y_t^i))\big) Z_{t+1}^i G^{i\prime}(y_t^i)$ is the discounted marginal value of one more unit of escapement of species i.

The implicit equation above results from differentiating the Bellman equation and using the fisherman's condition $y_t^i = (C^i)^{-1}(p^i - m\pi_t)$ (see (44) in Appendix B.2). This Euler equation balances the current marginal profit with the weighted average of next-period marginal profits.

Corollary 1.1 (The optimal $m\pi_t$ is constant)**.** *The optimal marginal profit is constant over time.*

Proof. Because the shocks are i.i.d. and Z_{t+1}^i is independent of y_t^i, the expectation $\mathbb{E}[\pi_{t+1}^i]$ is a deterministic function of y_t^i alone:

$$\pi_{t+1}^i(y_t^i) := \beta\, \mathbb{E}\Big[\big(p^i - C^i(Z_{t+1}^i G^i(y_t^i))\big) Z_{t+1}^i\Big] G^{i\prime}(y_t^i)$$

Substituting $y_t^i = Y^i(m\pi_t)$, Equation (13) becomes

$$m\pi_t = \sum_{i=1}^{N} \rho^i(m\pi_t)\, \pi^i_{t+1}\big(Y^i(m\pi_t)\big),$$

an equation in $m\pi_t$ alone that contains no state variable and no time index. Its solution (unique under Corollary 2.1) is therefore the same in every period, and $m\pi_t$ does not depend on $\vec{X}_t$. □

From this result, I obtain the following remark:

Remark 1.1 (Deterministic case)**.** *If $Z^i_t \equiv 1$, then $m\pi_t = m\pi_{t+1} = m\pi$.*

Proof. With $Z^i_t \equiv 1$, the expectation in (13) is degenerate and the argument of Corollary 1.1 applies directly. □

Under a specific stock-dependent cost function, the optimal margin is unaffected by the ecological shocks.

Corollary 1.2 (Specific cost function recovers the deterministic case)**.** *If the cost function is $C^i(s^i) = \frac{\theta^i}{s^i}$, the optimal margin under uncertainty equals the optimal margin under certainty.*

Proof. With $C^i(s) = \theta^i/s$ and $\mathbb{E}[Z^i_{t+1}] = 1$,

$$\mathbb{E}\Big[\Big(p^i - \frac{\theta^i}{Z^i_{t+1}G^i(y^i_t)}\Big)Z^i_{t+1}\Big]G^{i\prime}(y^i_t) = \Big(p^i - \frac{\theta^i}{G^i(y^i_t)}\Big)G^{i\prime}(y^i_t) = \big(p^i - C^i(G^i(y^i_t))\big)G^{i\prime}(y^i_t),$$

so π^i_{t+1} in the proof of Corollary 1.1 is the deterministic marginal value, and (13) coincides with its deterministic counterpart. With $C^i(s) = \theta^i/s^\alpha$ and $0 < \alpha < 1$, the shock no longer cancels and the deterministic margin is only a lower bound on the optimal one (see Appendix A). □

Remark 1.2 (Re-expressing the weights ρ^i_t)**.** *For a cost function $C^i(s^i) = \frac{\theta^i}{s^i}$, the weights ρ^i_t can be re-expressed as a function of the escapements and cost parameters: $\rho^i = \frac{(y^i)^2/\theta^i}{\sum_j (y^j)^2/\theta^j}$.*

Proof. The re-expression follows from using the cost function $C^i(s^i) = \frac{\theta^i}{s^i}$ and the fisherman's optimality condition $p^i - m\pi_t = C^i(y^i_t)$ in (13) (see expressions (45) and (46)). □

By inspection, the tax basket equals $m\pi^+_t$, which does not depend on the state variables, while the quota depends on the states through a stochastic shadow value. Consequently, the fisherman's escapement is deterministic under a tax ($\vec{\bar{y}}$) and stochastic under a quota ($\vec{y}_t$). In a deterministic setting, however, the shadow value is also deterministic, because $\lambda(\vec{X}_t, \omega_t) = \lambda(\vec{G}(\vec{y}_{t-1}), \omega_t)$. The shadow value and the tax basket then behave the same way: $\tau_t = \lambda(\omega_t)$. In addition, if all species have the same marginal profit at their respective optimal escapements, the basket instruments are first-best:

Corollary 1.3 (First-best basket instruments)**.** *Given the core assumptions, the tax basket implements the first best if and only if* $p^i - C^i(y^{i*}) = p^j - C^j(y^{j*})$ *for all* $i \neq j$*, that is, if all species-specific optimal taxes coincide. In a deterministic setting, the same condition makes the quota basket first-best.*

Proof. If species have identical marginal profits at their respective optimal escapements, Reed's optimality condition, which under $C^i = \theta^i/s$ and $\mathbb{E}[Z^i] = 1$ reads $\beta\big(p^i - C^i(G^i(y^{i*}))\big)G^{i\prime}(y^{i*}) = p^i - C^i(y^{i*})$, holds for every species at the common margin:

$$\begin{aligned} p^1 - C^1\left(y_t^1\right) = p^2 - C^2\left(y_t^2\right) = \cdots = p^N - C^N\left(y_t^N\right) = \tau_t = \lambda(\omega_t) \\ p^1 - C^1\left(y^{1*}\right) = p^2 - C^2\left(y^{2*}\right) = \cdots = p^N - C^N\left(y^{N*}\right) = \tau^* = \lambda(\omega_t) \\ \beta\big(p^1 - C^1(G^1(y^{1*}))\big)G^{1\prime}(y^{1*}) = \cdots = \beta\big(p^N - C^N(G^N(y^{N*}))\big)G^{N\prime}(y^{N*}) = \tau^* = \lambda(\omega_t) \end{aligned}$$

Even if species' profits are heterogeneous, a fixed tax or shadow value then achieves Reed's constant escapement for every species. Because the tax does not depend on the shocks, the result for the tax basket also holds under uncertainty. □

This analysis is incomplete: I still need to prove formally that $\tau^+ = m\pi_t^+$, and to show whether choosing the optimal shadow value is equivalent to choosing the optimal quota ω_t^+ that delivers $m\pi^+$. The next subsection proves both.

2.3 Planner's problem

After studying the fisherman's decision and recognizing how the dynamic choice of $m\pi^+$ shapes the harvest decision, the planner chooses the level of the tax or quota basket to maximize intertemporal profits. For the tax basket, the solution of the dynamic problem is straightforward because it follows the same procedure as in Appendix B after replacing $m\pi_t = \tau_t$. Because the tax is independent of the state variables in both the fisherman's and the planner's problems, it satisfies the same optimality condition as the welfare-maximizing marginal profit, Equation (13) in Proposition 1.

Proposition 2 (Welfare-maximizing tax basket)**.** *Given the assumptions for* N *heterogeneous and independent stocks, species-specific optimal taxes* τ^{i*}*, and the interior regularity condition*

$$\max_i \tau^{i*} \;<\; \tau^F_{\max} \;:=\; \min_i\big(p^i - C^i(K^i)\big),$$

which keeps all species in the fishable set on $[\min_i \tau^{i*}, \max_i \tau^{i*}]$*, the welfare-maximizing tax basket* τ^+ *is characterized by the same implicit equation as in Proposition 1:*

$$\tau_t^+ = \mathbb{E}\Big[\sum_{i=1}^{N} \underbrace{\rho_t^i(\tau_t^+)}_{\text{Weight } i} \underbrace{\pi_{t+1}^i}_{\text{Marginal profit } i \text{ at } t+1}\Big] \tag{14}$$

where $\rho_t^i = \frac{y^i_{\tau_t^+}}{\sum_{j=1}^N y^j_{\tau_t^+}}$ *is the per-species weight, and* $\pi^i_{t+1}(\tau^+) = \beta\big(p^i - C^i(Z^i_{t+1}G^i(y^i_t))\big)Z^i_{t+1}G^{i\prime}(y^i_t)$ *is the next-period marginal profit.*

Welfare is strictly lower than under species-specific first-best taxes whenever the τ^{i*} *are not all equal;* τ^+ *is then second-best. The tax basket* τ^+ *also satisfies Corollaries 1.1 to 1.3.*

Proof. Replacing $m\pi_t^+ = \tau_t^+$ in Proposition 1 and its corollaries shows that the optimal tax basket satisfies the same Euler equation as the optimal marginal profit and is fixed over time (for a complete derivation, see Appendix B.3). □

Decomposing the per-period profits,

$$\int_{y_t^i}^{X_t^i}\left(p^i - \frac{\theta^i}{s}\right)ds \;=\; p^iX_t^i - \theta^i \ln X_t^i - p^iy_t^i + \theta^i \ln y_t^i, \tag{15}$$

substituting $X_t^i = Z_t^iG^i(y_{t-1}^i)$, and using $\mathbb{E}[Z_t^i] = 1$ and $\mathbb{E}|\ln Z_t^i| < \infty$, I recover Proposition 2: the same problem repeats each period, and it has the same solution even in a stochastic setting. If the condition of Corollary 1.3 holds, the tax basket is first-best because all individual optimal taxes are equal. Otherwise, the tax basket is a second-best policy because it implements a constant escapement per species that differs from its individual Reed escapement. It maximizes profits with a single instrument, but it does not achieve the outcome of species-specific management.

Corollary 2.1 (Existence, uniqueness and interiority of τ^+)**.** *Under Proposition 2, the regularity condition* $\max_i \tau^{i*} < \tau^F_{\max} := \min_i(p^i - C^i(K^i))$*, the cost function* $C^i = \theta^i/s^i$*, and the second-order condition*

$$y\,\pi^{i\prime}_{t+1}(y) + 2\pi^i_{t+1}(y) \;\le\; p^i + \tau, \tag{16}$$

The optimal tax basket exists, is unique, and is interior.

The full proof is in Appendix D. Equation (16) (the same as (52)) is a second-order condition that guarantees global concavity and complements the fishable-set assumption to deliver a unique optimal tax basket. The fishable-set condition keeps every species harvested as τ varies, and (16) keeps each species' contribution concave. Together, they guarantee that a basket is not too heterogeneous, which formalizes the similarity condition for species within a bundle (Collado et al., 2021, 2024; Costello and Szuwalski, 2018; Kritzer et al., 2023). Taken together, these results show that tax baskets preserve the constant-escapement structure under uncertainty but generally implement a second-best allocation when species differ in marginal profits. Consequently, τ^+ is the peak of basket welfare as a function of the margin (Section 2.4).

First-best outcomes arise only when marginal profits are identical across species at their respective optima. For the policymaker, data on the economic and ecological parameters are

enough to determine the optimal tax basket. However, the planner needs current stock data if, in some period, a species' realized stock falls below its target escapement, $X_t^i \leq Y^i(\tau^+)$, so that the species drops out of the harvest. In that case, the interiority condition breaks down.

In contrast, the quota basket requires more information because it changes over time. The essential difference from the tax basket is that a single quota constrains only aggregate harvest, so the split across species is the fisherman's choice. Thus, the shadow value λ_t is not a policy instrument but an equilibrium object that depends on the current stocks and the quota, $\lambda_t = \lambda(\vec{X}_t, \omega_t)$.

Proposition 3 (Welfare-maximizing quota basket)**.** *Under the setup of Proposition 2, the welfare-maximizing quota ω^+ satisfies*

$$\mathbb{E}\Big[\underbrace{\lambda(\vec{X}_t, \omega_t)}_{\textit{Current shadow value}} - \beta \sum_i \rho_t^i \underbrace{V_{y^i}(\vec{y}_t)}_{\textit{Continuation value}}\Big] = 0, \tag{17}$$

where λ_t is the fisherman's shadow value, V is the continuation value function, and $\rho_t^i = \frac{Y_\lambda^i(\lambda_t)}{\sum_j Y_\lambda^j(\lambda_t)}$ is the per-species reallocation weight.

The derivation is in Appendix E.3. Expression (17) equates, in expectation, the current shadow value of an extra unit of aggregate harvest with the ρ-weighted continuation value of the escapement. By the envelope theorem, the marginal value of escapement of species k is the value of one extra fish in stock k, minus the shadow value λ_t of the escapement it displaces, plus its continuation value:

$$V_{y_{t-1}^k}(\vec{y}_{t-1}) = \mathbb{E}\Big[\Big(p^k - C^k(X_t^k) - \lambda_t + \beta \sum_j \rho_t^j V_{y^j}(\vec{y}_t)\Big) Z_t^k G^{k\prime}(y_{t-1}^k)\Big] \tag{18}$$

Separating the optimality condition into two parts is useful because (17) is the rule for the level of the quota basket, and (18) reveals what happens when the shadow value targeted by the planner, λ^+, differs from the effective shadow value the fisherman selects after the shocks, λ_t^F. Before discussing how the planner's inability to observe the ecological shocks causes a mismatch between these shadow values, I establish that choosing λ^+ is equivalent to choosing τ^+:

Corollary 3.1 (Duality of λ^+ and τ^+)**.** *Choosing the optimal quota ω_t^+ is equivalent to choosing an optimal shadow value λ^+, in the sense that the planner's first-order condition reduces to*

$$\lambda^+ = \beta \mathbb{E}\Big[\sum_{i=1}^N \big(p^i - C^i(Z_{t+1}^i G^i(y_t^i))\big) Z_{t+1}^i G^{i\prime}(y_t^i) \frac{y_{\lambda^+}^i}{\sum_{j=1}^N y_{\lambda^+}^j}\Big], \tag{19}$$

which is identical in form to the tax condition (14). Both instruments target the same escapement vector $\bar{y}^i = \theta^i/(p^i - \tau^+)$.

The equivalence is demonstrated in Appendix E. Thanks to duality, it is straightforward to establish that (i) λ^+ is fixed over time; (ii) λ^+ exists, is unique, and is interior under the same conditions as τ^+; and (iii) the quota ω_t is not constant. In other words, the two instruments coincide at the planner's optimum but differ in delivery. A planner who targets λ^+ sets the quota basket before the shocks, as the difference between the expected aggregate stock and the sum of target escapements:

$$\omega_t^+ = \sum_i G^i(y_{t-1}^i) - \sum_i Y^i(\lambda^+), \qquad \sum_i G^i(y_{t-1}^i) = \mathbb{E}_{t-1}\Big[\sum_i X_t^i\Big]. \tag{20}$$

If the planner could observe the realized stocks X_t^i instead, the quota $\sum_i X_t^i - \sum_i Y^i(\lambda^+)$ would induce $\lambda_t^F = \lambda^+$ in every state, and the two instruments would coincide. I refer to (20) as the certainty-equivalent quota.

As with a tax basket, determining the level of a quota basket does not require knowing the stock composition across species, but it does require knowing total biomass. By the end of period t, comparing a quota basket under complete information with one under asymmetric information shows the difference between the fisherman's shadow value λ_t^F and the planner's λ^+:

$$\sum_i (Z_t^i - 1)\, G^i(y_{t-1}^i) = \sum_i \Big[(C^i)^{-1}(p^i - \lambda_t^F) - (C^i)^{-1}(p^i - \lambda^+)\Big], \tag{21}$$

so λ_t^F inherits the randomness of the aggregate realized stock. The following corollary identifies the source of the distortion:

Corollary 3.2 (The distortion between τ^+ and ω_t)**.** *In a stochastic setting, the marginal value of escapement under the quota basket decomposes as*

$$\beta V_{y^k}(\vec{y}_{t-1}) = \pi_{t+1}^k(y_{t-1}^k) - \beta\,\mathrm{Cov}\big(R_t, Z_t^k\big)\, G^{k\prime}(y_{t-1}^k), \qquad R_t := \lambda_t^F - \beta \sum_j \rho_t^j V_{y^j}(\vec{y}_t), \tag{22}$$

where the distortion is the covariance between the residual of the planner's FOC, R_t, and the ecological shock of species k.

The full proof is in Appendix E.5. The covariance vanishes only in the deterministic case, where $\lambda_t^F = \lambda^+$ every period. When the species-specific optimal taxes coincide, the tax basket is first-best (Corollary 1.3) but the quota basket is not, because λ_t^F still moves with the realized shocks. The next subsection shows that, under uncertainty, the tax basket weakly outperforms the quota basket.

2.4 Welfare comparison

To compare the welfare outcomes across instruments, I use Equation (47) (see the details in Appendix C.1):

$$\mathcal{W} = \sum_i \underbrace{\left[A^i(X_0^i) - \kappa^i \tfrac{\beta}{1-\beta}\right]}_{=:\Xi_0} + \sum_{t\geq 0} \beta^t \sum_i \underbrace{\left[\phi^i(y_t^i) + \beta a^i(y_t^i)\right]}_{=:w^i(y_t^i)}$$

with $a^i(y) := p^i G^i(y) - \theta^i \ln G^i(y)$ and $\kappa^i := \theta^i \mathbb{E}[\ln Z_t^i]$. The decomposition holds in expectation under either instrument: the shock Z_{t+1}^i enters only through $\mathbb{E}\left[A^i(Z_{t+1}^i G^i(y_t^i)) \,\middle|\, y_t^i\right] = a^i(y_t^i) - \kappa^i$, and Z_{t+1}^i is independent of y_t^i because the shocks are weakly exogenous. Because Ξ_0 and the shock path are common to both instruments, they cancel in the difference. The per-species payoff satisfies

$$w^{i\prime}(y) = \pi_{t+1}^i(y) - \big(p^i - C^i(y)\big), \qquad w^{i\prime\prime}(y) = \phi^{i\prime\prime}(y) + \beta a^{i\prime\prime}(y) < 0 \tag{23}$$

Setting $w^{i\prime}(y) = 0$ gives $\pi_{t+1}^i(y) = p^i - C^i(y)$, which is Reed's condition for species i alone. Hence w^i peaks at the species-specific escapement y^{i*}, not at the basket escapement $\bar{y}^i = Y^i(\tau^+)$. Let $\mathcal{M} := [\lambda_L, \lambda_u]$ be an interval that contains τ^+ and every shadow value the quota can realize (see Appendix F), and define the per-period basket welfare as a function of the margin:

$$\mathcal{W}(m\pi_t) := \Psi(m\pi_t) = \sum_{i=1}^N \psi^i(m\pi_t), \qquad \psi^i(m\pi_t) := w^i\big(Y^i(m\pi_t)\big), \qquad m\pi_t \in \mathcal{M} \tag{24}$$

Under the tax, $m\pi_t = \tau^+$ is deterministic, so the objective separates into a deterministic sequence of identical static problems. Under the quota, $m\pi_t = \lambda_t^F = \lambda(\vec{X}_t, \omega_t)$ is random, and $\Psi(\lambda_t^F)$ inherits the randomness of the realized stocks. When the margin moves, individual species gain or lose at first order, because $w^{i\prime}(\bar{y}^i) \neq 0$ unless all species-specific optimal taxes coincide (Appendix G). These first-order effects cancel across species. By Appendix D, $\mathcal{W}'(\tau) = -\big(\sum_i Y^{i\prime}(\tau)\big) g(\tau)$, and $g(\tau^+) = 0$ is the first-order condition of Proposition 2, so $\sum_i w^{i\prime}(\bar{y}^i)\, Y^{i\prime}(\tau^+) = \mathcal{W}'(\tau^+) = 0$. The comparison between instruments is therefore decided at second order, by the curvature of $\mathcal{W}$.

Proposition 4 (The welfare is higher under a tax basket)**.** *Let the tax basket be τ^+ and let the quota ω_t follow any rule that uses only information available before the shocks of period t, including the certainty-equivalent rule (20) and the exact optimum. Under the interior regularity condition, i.i.d. shocks with $\mathbb{E}[Z_t^i] = 1$, and condition (52) on $\mathcal{M} \cup [\tau^{1*}, \tau^{N*}]$,*

$$\mathbb{E}[\mathcal{W}^\tau] \;\geq\; \mathbb{E}[\mathcal{W}^\omega],$$

with strict inequality whenever $\Pr(\lambda_t^F \neq \tau^+) > 0$ for some t.

Proof. Because welfare is a function of the margin, with Ξ_0 and the shock path common to both instruments,

$$\mathbb{E}[\mathcal{W}^{\tau}] - \mathbb{E}[\mathcal{W}^{\omega}] = \sum_{t \geq 0} \beta^t \, \mathbb{E}\big[\Psi(\tau^+) - \Psi(\lambda_t^F)\big]$$

Because $\Psi'(\tau^+) = 0$ and condition (52) makes Ψ strictly concave on $\mathcal{M}$ (Corollary 2.1), $\Psi(\tau^+) - \Psi(m\pi) \geq 0$ for every $m\pi \in \mathcal{M}$, with equality only at $m\pi = \tau^+$. Since $\lambda_t^F \in \mathcal{M}$, the inequality $\Psi(\lambda_t^F) \leq \Psi(\tau^+)$ holds in every period and every state of nature, not only on average. Taking expectations of these nonnegative differences and summing them with positive weights β^t gives $\mathbb{E}[\mathcal{W}^{\tau}] - \mathbb{E}[\mathcal{W}^{\omega}] \geq 0$.

If $\Pr(\lambda_t^F \neq \tau^+) > 0$ for some t, the welfare integrand is strictly positive with positive probability, so that term, and the sum, are strictly positive. Under the certainty-equivalent rule, $\lambda_t^F = \tau^+$ if and only if the aggregate surprise S_t defined in (25) is zero, which has probability zero for continuously distributed, non-degenerate shocks. □

The argument never uses how the quota is set: any cap chosen before the shocks places the economy at $\Psi(\lambda_t^F)$, while the tax places it at the peak $\Psi(\tau^+)$ every period. Equivalently, by Appendix E.5, the tax attains $\beta V^{\tau}_{y^k} = \pi^k_{t+1}$ exactly, while the quota carries the distortion $-\beta \operatorname{Cov}(R_t, Z_t^k)\, G^{k\prime}$.

Under the assumptions of this paper, Weitzman's (2002) result holds: a tax outperforms a quota even under bundled management. This outcome rests on the cost function because the tax is stock-independent. Appendix B.3 shows that, with decreasing catchability, the deterministic tax is only a lower bound, so the tax is no longer immune to shocks; I conjecture that the ranking then narrows. Adding species to the basket also narrows the gap per species, although the total gap settles at a positive constant. Before proving this, I collect the bounds that the proof requires:

Assumption 1 (Uniform bounds)**.** *There are constants, independent of N, such that for every species i and every margin $m\pi \in \mathcal{M} = [\lambda_L, \lambda_u]$:*

(a) price and cost parameter: $p^i \in [\underline{p}, \overline{p}]$ and $\theta^i \in [\underline{\theta}, \overline{\theta}]$, with $\lambda_u < \underline{p}$;

(b) growth: $\underline{G} \leq G^i(y) \leq \overline{G}$ for every escapement $y \geq Y^i(\lambda_L)$;

(c) curvature: $\underline{c} \leq -\psi^{i\prime\prime}(m\pi) \leq \overline{c}$, with $0 < \underline{c} \leq \overline{c} < \infty$;

(d) shocks: $\{Z_t^i\}_i$ are independent across species, with $\operatorname{Var}(Z_t^i) = \sigma_Z^2 > 0$.

Parts (a) and (b) keep any single species from dominating the basket as N grows. Part (c) is a uniform, strict version of condition (16). Part (d) is where independence across species enters. With this assumption, I can evaluate how diversification affects the uncertainty loss:

Proposition 5 (Diversification of uncertainty loss)**.** *Let the quota basket follow the certainty-equivalent rule (20), with X_0^i known when ω_0 is set (for simplicity). Under Assumption 1 and the conditions of Proposition 4, define*

$$\underline{\ell} := \frac{\underline{c}\,\sigma_Z^2\,\underline{G}^{\,2}}{2\,\overline{m}^{\,2}}, \qquad \overline{\ell} := \frac{\overline{c}\,\sigma_Z^2\,\overline{G}^{\,2}}{2\,\underline{m}^{\,2}},$$

with the slope bounds $\underline{m}, \overline{m}$ of Lemma 2. Then, for every basket size N,

$$\frac{\beta}{1-\beta}\,\underline{\ell} \;\le\; \mathbb{E}[\mathcal{W}^{\tau}] - \mathbb{E}[\mathcal{W}^{\omega}] \;\le\; \frac{\beta}{1-\beta}\,\overline{\ell}$$

Consequently, the average per-species gap $\big(\mathbb{E}[\mathcal{W}^{\tau}] - \mathbb{E}[\mathcal{W}^{\omega}]\big)/N$ is of order $1/N$ and vanishes as $N \to \infty$, while the total gap stays bounded away from zero. The upper bound also holds when the quota is set optimally.

Proof. Let $\mathcal{H}_{t-1}$ denote the information available when the planner sets ω_t: it contains $\vec{y}_{t-1}$. Then, the certainty-equivalent quota is $\omega_t = \sum_i G^i(y_{t-1}^i) - \sum_i Y^i(\tau^+)$. The fisherman's budget is $\sum_i Y^i(\lambda_t^F) = \sum_i X_t^i - \omega_t$. Subtracting,

$$\begin{aligned} \sum_i Y^i(\lambda_t^F) - \sum_i Y^i(\tau^+) &= \sum_i X_t^i - \sum_i G^i(y_{t-1}^i) \\ \sum_i \big[Y^i(\lambda_t^F) - Y^i(\tau^+)\big] &= \sum_i (Z_t^i - 1)\, G^i(y_{t-1}^i) =: S_t \end{aligned} \tag{25}$$

Conditional on $\mathcal{H}_{t-1}$, the terms $G^i(y_{t-1}^i)$ are fixed, and the Z_t^i are independent across i with mean one:

$$\begin{aligned} \mathbb{E}[S_t \mid \mathcal{H}_{t-1}] &= \sum_i \mathbb{E}[Z_t^i - 1]\, G^i(y_{t-1}^i) = 0 \\ \mathbb{E}[S_t^2 \mid \mathcal{H}_{t-1}] &= \sum_i \sum_j \mathbb{E}\big[(Z_t^i - 1)(Z_t^j - 1)\big] G^i(y_{t-1}^i) G^j(y_{t-1}^j) \\ &= \sum_i \sigma_Z^2 \left[G^i(y_{t-1}^i)\right]^2 \qquad \text{cross terms vanish} \end{aligned} \tag{26}$$

Every realized escapement satisfies $y_{t-1}^i = Y^i(\lambda_{t-1}^F) \ge Y^i(\lambda_L)$, so Assumption 1(b) applies to each term. By iterated expectations,

$$N\sigma_Z^2\,\underline{G}^{\,2} \;\le\; \mathbb{E}[S_t^2] \;\le\; N\sigma_Z^2\,\overline{G}^{\,2} \tag{27}$$

The variance of the aggregate surprise grows linearly in N (law-of-large-numbers step). By the mean value theorem, each term in (25) equals $Y^{i\prime}(\zeta_t^i)(\lambda_t^F - \tau^+)$ for some $\zeta_t^i \in \mathcal{M}$

between λ_t^F and τ^+. Every Y^i is strictly increasing, so all N terms share the sign of $\lambda_t^F - \tau^+$ and reinforce rather than cancel. With the slope bounds of Lemma 2,

$$N\underline{m}\,|\lambda_t^F - \tau^+| \;\le\; |S_t| \;\le\; N\overline{m}\,|\lambda_t^F - \tau^+|$$
$$\frac{S_t^2}{N^2\overline{m}^{\,2}} \;\le\; (\lambda_t^F - \tau^+)^2 \;\le\; \frac{S_t^2}{N^2\underline{m}^{\,2}}$$

Taking expectations and using (27),

$$\frac{\sigma_Z^2\,\underline{G}^{\,2}}{N\,\overline{m}^{\,2}} \;\le\; \mathbb{E}\big[(\lambda_t^F - \tau^+)^2\big] \;\le\; \frac{\sigma_Z^2\,\overline{G}^{\,2}}{N\,\underline{m}^{\,2}} \tag{28}$$

The margin's mean squared deviation falls like $1/N$. A surprise of size $\sqrt{N}$ is spread over N species. Since $\Psi = \sum_i \psi^i$, Assumption 1(c) gives, for every $m\pi \in \mathcal{M}$,

$$N\underline{c} \;\le\; -\Psi''(m\pi) \;\le\; N\overline{c}$$

The first-order condition of Proposition 2 gives $\Psi'(\tau^+) = 0$ (Appendix D), so Lemma 3 applies with the curvature bounds:

$$\frac{N\underline{c}}{2}\,(\lambda_t^F - \tau^+)^2 \;\le\; \Psi(\tau^+) - \Psi(\lambda_t^F) \;\le\; \frac{N\overline{c}}{2}\,(\lambda_t^F - \tau^+)^2$$

This is where the species-level first-order terms of Lemma 4 cancel: they add up to $\Psi'(\tau^+)(\lambda_t^F - \tau^+) = 0$. Taking expectations in this inequality and using (28),

$$\mathbb{E}\big[\Psi(\tau^+) - \Psi(\lambda_t^F)\big] \le \frac{N\overline{c}}{2} \cdot \frac{\sigma_Z^2\,\overline{G}^{\,2}}{N\,\underline{m}^{\,2}} = \frac{\overline{c}\,\sigma_Z^2\,\overline{G}^{\,2}}{2\,\underline{m}^{\,2}} = \overline{\ell}$$
$$\mathbb{E}\big[\Psi(\tau^+) - \Psi(\lambda_t^F)\big] \ge \frac{N\underline{c}}{2} \cdot \frac{\sigma_Z^2\,\underline{G}^{\,2}}{N\,\overline{m}^{\,2}} = \frac{\underline{c}\,\sigma_Z^2\,\underline{G}^{\,2}}{2\,\overline{m}^{\,2}} = \underline{\ell}$$

Curvature of order N times a squared deviation of order $1/N$ gives a per-period loss of order one, with constants that do not depend on N. At $t = 0$ the stocks X_0^i are known, so $\omega_0 = \sum_i X_0^i - \sum_i Y^i(\tau^+)$ induces $\lambda_0^F = \tau^+$ and the period-0 loss is zero. By Proposition 4,

$$\mathbb{E}[\mathcal{W}^\tau] - \mathbb{E}[\mathcal{W}^\omega] = \sum_{t\ge 1} \beta^t\,\mathbb{E}\big[\Psi(\tau^+) - \Psi(\lambda_t^F)\big]$$
$$\frac{\beta}{1-\beta}\,\underline{\ell} \;\le\; \mathbb{E}[\mathcal{W}^\tau] - \mathbb{E}[\mathcal{W}^\omega] \le \frac{\beta}{1-\beta}\,\overline{\ell}$$

Dividing by N, the average per-species gap lies between $\frac{\beta\underline{\ell}}{(1-\beta)N}$ and $\frac{\beta\overline{\ell}}{(1-\beta)N}$, so it is of order $1/N$ and vanishes, while the total gap stays above $\frac{\beta\underline{\ell}}{1-\beta} > 0$. Finally, the optimal quota ω_t^* does at least as well as the certainty-equivalent quota, so, by Proposition 4,

$$0 \;\le\; \mathbb{E}[\mathcal{W}^\tau] - \mathbb{E}[\mathcal{W}^{\omega^*}] \;\le\; \mathbb{E}[\mathcal{W}^\tau] - \mathbb{E}[\mathcal{W}^\omega] \;\le\; \frac{\beta}{1-\beta}\,\overline{\ell}$$

□

The escapements also converge: by Lemma 2 and (28), $\mathbb{E}\big[(y_t^i - \bar{y}^i)^2\big] \le \overline{m}^2\,\mathbb{E}\big[(\lambda_t^F - \tau^+)^2\big] = O(1/N)$, so the quota's realized escapement vector approaches the tax's target $\vec{\bar{y}}$ species by species. The result also concerns the total and the average: an individual species can do better under the quota in a given period (Lemma 4). Adding species brings the quota basket's average per-species outcome closer to the tax basket, but the total gap does not disappear. The result fails when ecological shocks are correlated across species:

Remark 5.1 (Correlated shocks)**.** *Independence across species (Assumption 1(d)) is essential. If* $\mathrm{Corr}(Z_t^i, Z_t^j) = \eta_Z > 0$ *for* $i \neq j$*, the cross terms in (26) no longer vanish and* $\mathbb{E}[S_t^2] \ge \sigma_Z^2\left[N + \eta_Z N(N-1)\right]\underline{G}^2$*, which grows with* N^2*. Thus, the per-species gap no longer vanishes; it converges to a positive floor (Section 3.3.3).*

As in portfolio theory, only idiosyncratic risk diversifies. System-wide shocks, such as climate change or El Niño events, weaken quota baskets' ability to diversify ecological risk. Correlation does not affect the tax basket, which remains the welfare-preferred instrument for every basket size. However, as N increases, both basket instruments incur an efficiency loss relative to species-specific management:

Proposition 6 (Comparison between the first best and basket instruments)**.** *The welfare of managing the species individually (a species-specific first-best tax* τ^{i*} *for each stock) is at least as large as the welfare of managing the same species under a single tax basket* τ^+*, and strictly larger unless all species share the same optimal tax.*

Proof. From the welfare decomposition (47), both regimes share the policy-independent constant $\Xi_0 = \sum_i [A^i(X_0^i) - \kappa^i\beta/(1-\beta)]$, and each reduces to a discounted stream of the dynamic per-species payoff $w^i(y) = \phi^i(y) + \beta a^i(y)$. Since escapement is constant over time under a fixed tax, the streams collapse to a $\frac{1}{1-\beta}$ factor, and the difference is

$$\mathcal{W}^{\mathrm{ind}} - \mathcal{W}^{\tau} = \frac{1}{1-\beta}\sum_{i=1}^{N}\Big[\, w^i\big(Y^i(\tau^{i*})\big) - w^i\big(Y^i(\tau^+)\big)\,\Big], \tag{29}$$

where Ξ_0 has cancelled. By (23), $w^{i\prime}(y) = \pi_{t+1}^i(y) - (p^i - C^i(y))$, which vanishes exactly at the individual optimum: $Y^i(\tau^{i*})$ satisfies Reed's condition. Since w^i is strictly concave, $Y^i(\tau^{i*})$ is its unique global maximizer, so every bracket in (29) is nonnegative:

$$w^i\big(Y^i(\tau^{i*})\big) \ge w^i\big(Y^i(\tau^+)\big) \qquad \text{for all } i,$$

strictly whenever $\tau^+ \neq \tau^{i*}$. Summing and dividing by $1-\beta > 0$ gives $\mathcal{W}^{\mathrm{ind}} - \mathcal{W}^{\tau} \ge 0$, with equality only when $\tau^{i*} = \tau^+$ for every i. Combined with Proposition 4 ($\mathcal{W}^{\tau} \ge \mathcal{W}^{\omega}$), this orders the three regimes:

$$\mathcal{W}^{\text{ind}} \;\geq\; \mathcal{W}^{\tau} \;\geq\; \mathcal{W}^{\omega}$$

□

The individual tax places each species at the peak of its own dynamic payoff w^i. The basket forces a single τ^+ on every species, pushing each off-target stock down its own concave hill. This shortfall is the mismatch loss in (5).

3 Numerical results

To verify the welfare comparison, I simulate welfare under each instrument using the same randomly generated fisheries and the same Monte Carlo shocks. I use simulated rather than empirical fisheries so that every basket satisfies the regularity condition by construction.

3.1 The dataset

Each replication draws a basket of N species that satisfies the regularity condition (12) under the cost function $C^i = \theta^i / s^i$. Drawing $(r^i, K^i, p^i, \theta^i)$ naively becomes impractical as N grows because the rejection rate explodes, which illustrates that species in a basket cannot be too heterogeneous (Corollary 2.1). To avoid this, I fix each species' Reed escapement as a fraction of its carrying capacity, $\iota^i := y^{i*}/K^i$, and invert Reed's condition for the cost coefficient:

$$\theta^i \;=\; \frac{p^i\left(\frac{1}{\beta} - G^{i\prime}(y^{i*})\right)}{\dfrac{1}{\beta y^{i*}} - \dfrac{G^{i\prime}(y^{i*})}{G^i(y^{i*})}}, \qquad y^{i*} = \iota^i K^i. \tag{30}$$

Holding p^i and ι^i fixed, the optimal tax does not vary with K^i. I therefore draw the parameters first and then compute the cost coefficient θ^i. The ranges are narrow enough that no single species dominates the aggregate. Growth rates and carrying capacities capture biological heterogeneity, and prices capture economic heterogeneity.

Growth shocks Z_t^i are lognormal with unit mean, i.i.d. over time and, in the baseline, independent across species. They scale each stock's growth increment, $X_t^i = y_{t-1}^i + Z_t^i\, r^i y_{t-1}^i (1 - y_{t-1}^i/K^i)$. I set the baseline volatility to $\sigma_Z = 0.74$, where σ_Z is the standard deviation of $\ln Z_t^i$. This value is the mean standard deviation of log recruitment deviations estimated across exploited marine fish stocks by Thorson et al. (2014).

3.2 Monte Carlo simulation

For each $N \in \{2, 5, 10, 20, 30, 50\}$, I draw one basket and simulate all instruments over $T = 40$ periods on 2,000 shock paths (1,000 antithetic pairs). I implement the quota as a cap: the fisherman may land less than ω_t, in which case $\lambda_t^F = 0$.

I use two variance-reduction techniques: (i) common random numbers, so that the tax and the quota face identical shocks; and (ii) antithetic variates, which pair each path drawn with ε with a path drawn with $-\varepsilon$. Together they reduce the standard error of the estimated welfare gap. I verified that the tax and the quota yield identical welfare in a deterministic setting for heterogeneous baskets. Because shocks affect each instrument differently, I also report a sensitivity analysis for $\sigma_Z \in \{0.2, 0.4, 0.6, 0.74, 1.0\}$. Section 3.3.3 reports three robustness checks: a basket with one dominant species, species-specific volatilities, and shocks correlated across species.

I consider two basket scenarios: (1) a symmetric basket, in which all species have identical parameters but independent shocks; and (2) a heterogeneous (balanced) basket, in which species differ in parameters but no single species dominates (defined formally in Section 3.3.3). Welfare is measured in monetary units, as the expected present value over the $T = 40$ periods.

To decompose regulatory loss, I use two Monte Carlo layers: an inner layer that averages over ecological shock paths while holding a basket fixed, and an outer layer that averages over random baskets. This exercise averages 25 baskets, each simulated on 150 antithetic pairs (300 paths). I need the outer layer because the decomposition compares basket and species-specific instruments, and the latter vary considerably across baskets. Table 1 summarizes the parameters and simulation settings.

Parameter	Meaning	Range / value
r^i	intrinsic growth rate (per period)	$\mathcal{U}[0.30,\ 0.40]$
K^i	carrying capacity (biomass units)	$\mathcal{U}[1200,\ 2400]$
p^i	price (monetary units per biomass unit)	$\mathcal{U}[95,\ 105]$
$\iota^i = y^{i*}/K^i$	escapement fraction	$\mathcal{U}[0.48,\ 0.52]$
θ^i	cost coefficient	from (30)
β	discount factor	$1/1.1 \approx 0.909$
σ_Z	s.d. of $\ln Z_t^i$	$\{0.2, 0.4, 0.6, \mathbf{0.74}, 1.0\}$
T	horizon (periods)	40
N	basket size	$\{2, 5, 10, 20, 30, 50\}$
Paths (convergence)	paths per basket	2,000 (1,000 pairs)
Outer layer (decomposition)	baskets per N	25
Inner layer (decomposition)	paths per basket	300 (150 pairs)
Robustness checks (Section 3.3.3)		
Dominant species	share of aggregate biomass	0.6
σ_Z^i	species-specific s.d. of $\ln Z_t^i$	$\mathcal{U}[0.40,\ 1.08]$
	draws of $(\sigma_Z^i)_i$ per N; paths per draw	10; 400
η	correlation of $\ln Z_t^i$ across species	$\{\mathbf{0}, 0.2, 0.5, 0.8\}$
η_Z	implied correlation of Z_t^i	$\{0, 0.16, 0.43, 0.75\}$
	paths per (N, η) cell	2,000 (1,000 pairs)

Table 1: Parameters of the simulated fisheries. Initial stocks are set to the target escapement, $X_0^i = \bar{y}^i$, so both instruments start from the same state. Baseline values are in bold.

The fishable-set condition holds for the tax basket in every simulated period. Under the quota basket, ex-post shocks occasionally leave a stock below the escapement implied by the realized shadow value, $X_t^i < Y^i(\lambda_t^F)$. The fisherman then lands none of that species in that period. This corner reflects the quota's response to shocks rather than a change in the basket's composition. At baseline volatility, it occurs in 7% of species-periods at $N = 2$ and 1.4% at $N = 50$ in the heterogeneous basket, and in 8–14% in the dominant-species

scenario of Section 3.3.3. Corners become rarer as N grows beyond five because λ_t^F becomes more stable. This stabilization requires independent shocks: when shocks are correlated across species (Section 3.3.3), corners still occur in 6–8% of species-periods at $N = 50$.

3.3 Results

Panel (a) of Figure 2 shows that the expected welfare difference $\mathbb{E}[\mathcal{W}^\tau] - \mathbb{E}[\mathcal{W}^\omega]$ is positive at every N and settles at a positive constant. In the symmetric basket, it is 4,054 monetary units at $N = 2$ and 3,567 at $N = 50$. In the heterogeneous basket, it lies between about 3,300 and 3,650 for $N \geq 10$. Every estimate lies between 49 and 73 standard errors from zero. In relative terms, the quota's loss is 1.9% of the tax basket's welfare at $N = 2$, 0.35% at $N = 10$, and 0.07% at $N = 50$. The quota never outperforms the tax basket, and increasing N drives the total gap to a positive, stable constant.

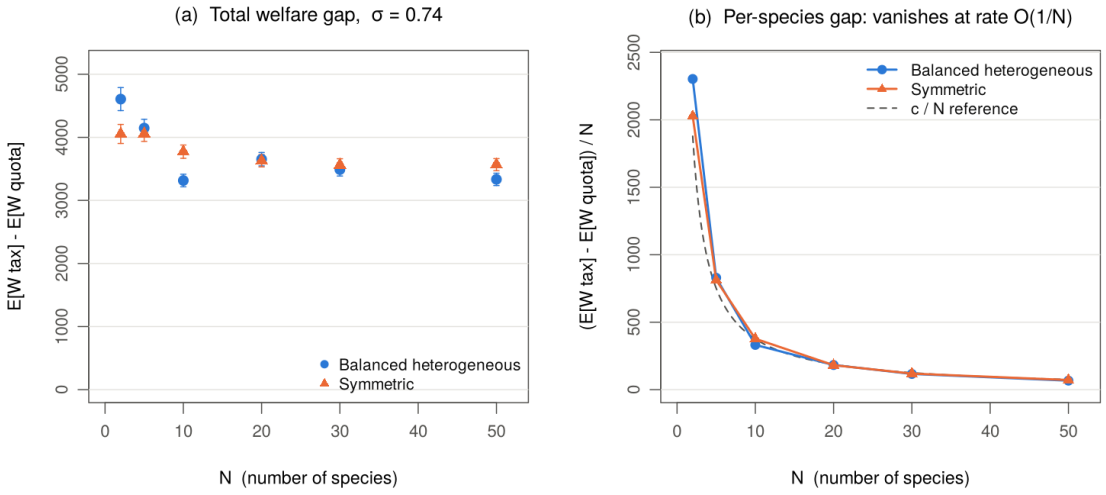


Figure 2: Expected welfare difference $\mathbb{E}[\mathcal{W}^\tau] - \mathbb{E}[\mathcal{W}^\omega]$ at $\sigma_Z = 0.74$. One basket per N, 2,000 shock paths. The bars are ± 1.96 standard errors. Panel (a): the total difference converges to a positive constant. Panel (b): the per-species difference vanishes at rate $O(1/N)$.

Panel (b) divides the result in panel (a) by N: the per-species difference vanishes at rate $O(1/N)$, following the c/N reference curve. This pattern is consistent with the quota's realized escapements converging to the tax basket's targets $\vec{y}$ as N grows (Proposition 5). Increasing N reduces the per-species gap in both the symmetric and the heterogeneous basket.

3.3.1 Sensitivity to shock volatility

The magnitude of the gap is governed by the variance of the shocks. Figure 3(b) shows that the total welfare difference rises steeply with volatility. Moving from $\sigma_Z = 0.2$ to the baseline 0.74 raises the shock variance by a factor of about 17 and the welfare difference by a factor of 16–20. At $\sigma_Z = 1$, the difference is 36–62 times its value at $\sigma_Z = 0.2$. Panel (a) reports the gap relative to the tax basket's welfare. In a two-species basket, it ranges from 0.1% at $\sigma_Z = 0.2$ to 4.7% at $\sigma_Z = 1$; at $N = 50$, it ranges from 0.004% to 0.14%. For every volatility, the relative gap declines in proportion to $1/N$, because the total difference stays flat while welfare grows with the number of species. The small gaps at low volatility therefore understate the quota's cost at empirically relevant volatility.

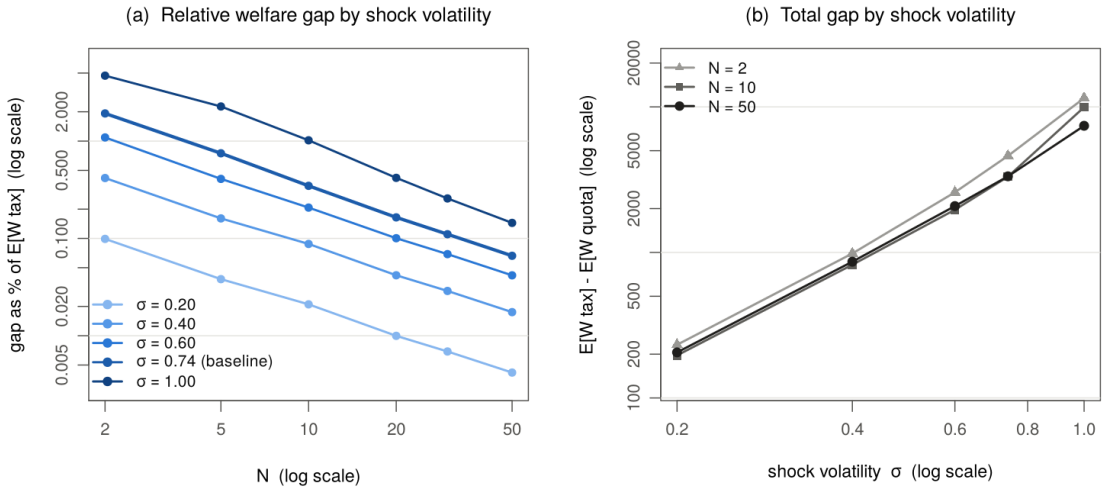


Figure 3: Sensitivity to shock volatility (heterogeneous basket). Panel (a): welfare difference as a percentage of $\mathbb{E}[\mathcal{W}^\tau]$, log–log axes. The baseline $\sigma_Z = 0.74$ is the thick line. Panel (b): total welfare difference against σ_Z for $N = 2, 10, 50$.

3.3.2 Decomposing the regulatory loss

Figure 4 shows that the two components of (5) move in opposite directions in N. The uncertainty loss is diversified away per species but not in total. It stays between 3,460 and 3,760 monetary units for $N \geq 5$ (4,428 at $N = 2$), and its per-species value falls from 2,214 at $N = 2$ to 70 at $N = 50$. The mismatch loss goes the opposite way. It grows roughly linearly, from 1,342 at $N = 2$ to 101,412 at $N = 50$, or 1.9% of first-best welfare.

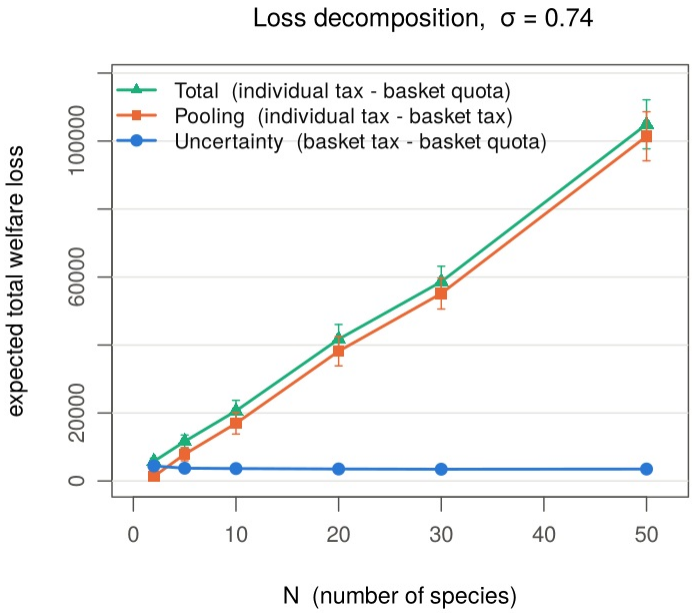


Figure 4: Decomposition of the total regulatory loss at $\sigma_Z = 0.74$, averaged over 25 heterogeneous (balanced) baskets per N (300 shock paths each). The bars show ± 1.96 standard errors across baskets.

The ranking of the two losses depends on basket size. In a two-species basket, the uncertainty loss is 3.3 times the mismatch loss (1.95% versus 0.59% of first-best welfare), so the choice between a tax and a quota is the first-order design decision. By $N = 5$, the mismatch loss is already twice the uncertainty loss, and by $N = 50$, it is 29 times larger. Using basket instruments therefore creates a trade-off between mismatch and uncertainty losses: the price–quantity choice matters most in small baskets, and the grouping decision dominates in large ones. This result extends previous research using groups of 2 (Clark, 1990), 5 (Collado et al., 2021), or 6 (Collado et al., 2024) species. These magnitudes likely overstate the mismatch loss. The benchmark $\mathcal{W}^{\text{ind}}$ assumes that species-specific management targets every stock perfectly and that stocks do not interact. In practice, choke species and ecological interactions reduce what species-specific management delivers, as in the West Coast groundfish fishery. The mismatch loss reported here is therefore best read as an upper bound on the cost of bundling.

3.3.3 Robustness

With the cost function $C^i(s^i) = \frac{\theta^i}{s^i}$, Remark 1.2 expressed the per-species weights ρ_t^i as a function of the escapements and cost parameters. Under a tax basket, each species reaches a constant escapement, and its weight also becomes constant. Under a quota basket, the

weights change every period with λ_t^F and stabilize only as $N \to \infty$. In both cases, the weights shrink as more species enter the basket. I define a basket as balanced when the largest weight also shrinks, $\max_i \rho^i \to 0$ as $N \to \infty$. What happens if one species dominates because it is highly valuable and abundant? Its weight then does not vanish, $\rho^i \to \bar{\rho} > 0$, and the variance of its escapement,[4] $\mathrm{Var}(y_t^i) \approx (\rho^i)^2 \mathrm{Var}(S_t) = \bar{\rho}^2\,\Theta(N)$, grows with N. In other words, a weight that does not shrink prevents the basket from spreading shocks across species.

To test this, I hold one species' share of aggregate biomass fixed at 60% as N grows, so its shock never diversifies away, and repeat the convergence exercise (one basket per N) with this dominant species. The per-species difference no longer vanishes. It stays between about 2,600 and 3,800 for $N \geq 5$, and the total difference grows from 3,165 at $N = 2$ to 147,014 at $N = 50$.

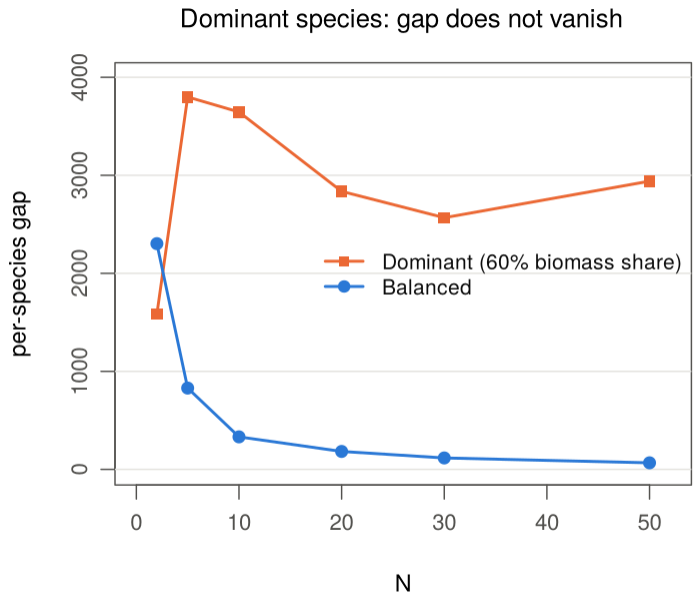


Figure 5: Robustness with one persistently dominant species. The per-species welfare difference does not vanish as basket size increases, unlike in a balanced basket.

This case shows that bundling does not work for every species composition. A successful basket is a balanced one, because it guarantees diversification. I conjecture that relaxing constant catchability would blur the tax basket's advantage over the quota but leave the mismatch loss intact. Appendix B.3 shows only that the tax condition then becomes a lower bound.

[4]Under the quota, a species absorbs the fraction ρ_t^i of the aggregate surprise, $y_t^i - \bar{y}^i \approx \rho_t^i S_t$, by (61).

A second check has each species draw its own volatility, $\sigma_Z^i \sim \mathcal{U}[0.40, 1.08]$. This range centers on the baseline 0.74 and spans about one predictive standard deviation (0.35) on each side in Thorson et al.'s stock-level estimates (2014). For each N, I keep the balanced basket of the main exercise and average over 10 draws of the volatility vector, with 400 shock paths per draw. The total welfare difference is 12–41% larger than with a common $\sigma_Z = 0.74$. Because the shock variance is convex in σ (for a lognormal shock with unit mean, $\mathrm{Var}(Z) = e^{\sigma^2} - 1$), spreading volatilities around 0.74 raises the average variance.

The per-species difference still declines at rate $1/N$, from 2,679 at $N = 2$ to 75 at $N = 50$ (panel (b) of Figure 6). Diversification therefore does not require species to share the same volatility.

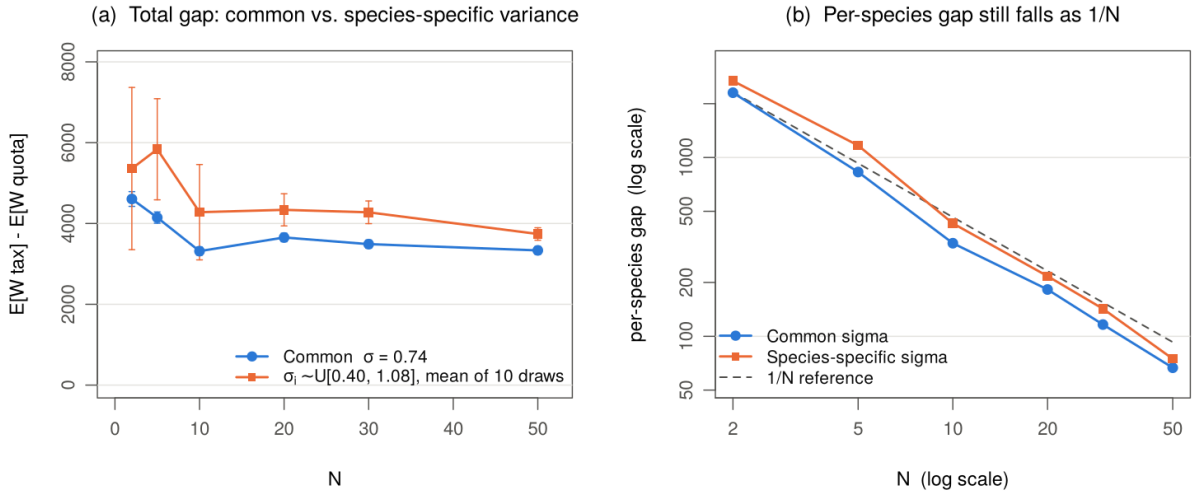


Figure 6: Robustness to species-specific volatilities, $\sigma_Z^i \sim \mathcal{U}[0.40, 1.08]$ (mean of 10 draws; the bars are ± 1.96 standard errors across draws), against a common $\sigma_Z = 0.74$ on the same basket. Panel (a): total welfare difference. Panel (b): per-species difference on log–log axes with a $1/N$ reference.

A third check relaxes independence across species. Species in a basket often share an environment, so a common driver such as temperature or upwelling can move their growth together. I let the log-shocks be equicorrelated,

$$\varepsilon_t^i = \sqrt{\eta}\, u_t + \sqrt{1-\eta}\, e_t^i, \qquad u_t,\, e_t^i \overset{iid}{\sim} \mathcal{N}(0,1), \tag{31}$$

with $Z_t^i = e^{\sigma_Z \varepsilon_t^i} / \mathbb{E}\left[e^{\sigma_Z \varepsilon_t^i}\right]$ as in the baseline, so that $\mathrm{corr}(\varepsilon_t^i, \varepsilon_t^j) = \eta$ for $i \neq j$, while each marginal, and therefore $\mathbb{E}[Z_t^i] = 1$, is unchanged. Because Z_t^i is lognormal, the correlation of the shocks themselves is $\eta_Z = (e^{\eta\sigma_Z^2} - 1)/(e^{\sigma_Z^2} - 1)$, which equals 0.16, 0.43 and 0.75 for $\eta = 0.2$, 0.5 and 0.8. Each stock still grows from its own escapement only, and because

the tax basket and the certainty-equivalent quota (20) depend on the shocks only through $\mathbb{E}[Z_t^i] = 1$, both instruments are set exactly as before. I rerun the heterogeneous (balanced) basket at $\sigma_Z = 0.74$ with 2,000 shock paths per cell.

Correlation weakens only the diversification of Proposition 5. With a common component, $\mathrm{Var}(S_t)$ grows with N^2 rather than N, and the variance of the realized shadow value tends to η_Z times its single-species value instead of zero, $\mathrm{Var}(\lambda_t^F) \propto 1/N + \eta_Z(1 - 1/N)$. Since the total difference is approximately N times a per-species loss proportional to $\mathrm{Var}(\lambda_t^F)$,

$$\mathbb{E}[\mathcal{W}^\tau] - \mathbb{E}[\mathcal{W}^\omega] \;\approx\; C\left[1 + \eta_Z(N-1)\right], \tag{32}$$

where C is the difference under independent shocks. The per-species difference therefore converges to $\eta_Z\, C$ rather than zero: a larger basket diversifies away only the idiosyncratic part of the risk.

Figure 7 confirms this. With $\eta = 0$, the results reproduce the main exercise. With correlated shocks, the total difference grows linearly in N and reaches 30,099, 75,323 and 124,314 monetary units at $N = 50$ for $\eta = 0.2$, 0.5 and 0.8. The per-species difference approaches a floor instead of vanishing: at $N = 50$ it is 602, 1,506 and 2,486, close to the limits $\eta_Z C$ implied by (32) (547, 1,489 and 2,600, with $C \approx 3{,}450$). For $N \geq 5$, (32) is within 13% of every simulated total, and within 5% at $N = 30$ and 50. In relative terms, the quota's loss at $N = 50$ is 0.6%, 1.5% and 2.5% of the tax basket's welfare, against 0.07% with independent shocks. The tax basket still dominates in every cell, with every estimate at least 48 standard errors from zero. The mismatch loss, by contrast, does not depend on the correlation (it varies by less than 0.5% across η), because it comes from the shared instrument rather than from the shocks.

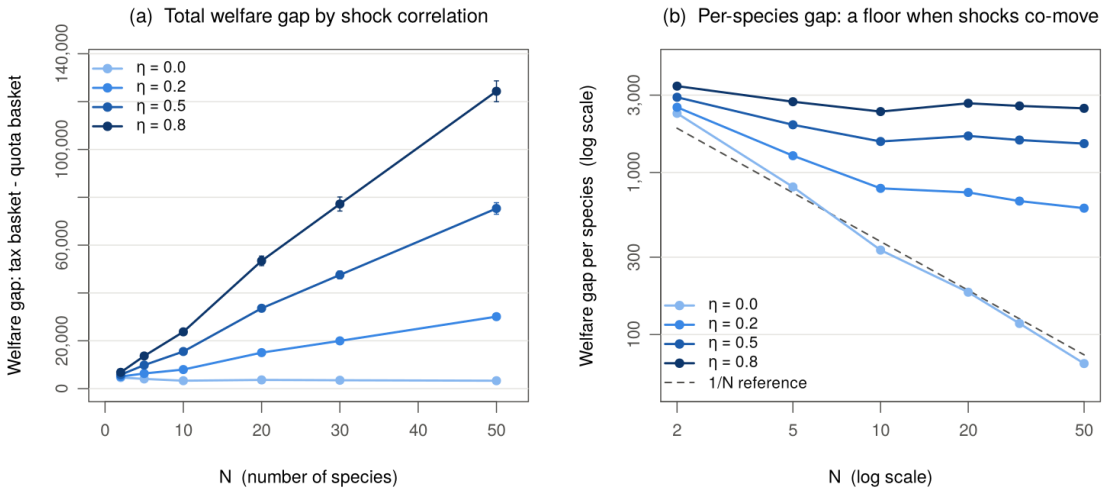


Figure 7: Robustness to shocks correlated across species, $\text{corr}(\ln Z_t^i, \ln Z_t^j) = \eta$ ($\sigma_Z = 0.74$, heterogeneous (balanced) basket, 2,000 shock paths per cell; the bars are ± 1.96 standard errors). Panel (a): total welfare difference. Panel (b): per-species difference on log–log axes with a $1/N$ reference; with $\eta > 0$, it approaches η_Z times the independent-shock total instead of vanishing.

A common shock therefore acts like a dominant species: in both cases, the whole basket shares part of the risk and cannot diversify it away. The more the species in a basket share an environment, the larger the quota's disadvantage and the stronger the case for the tax basket.

4 Discussion and conclusion

This is the first paper to study the incentives and trade-offs of basket instruments for heterogeneous stocks under ecological uncertainty. Building on and extending the seminal work of Reed (1979) and Weitzman (2002), I find that tax baskets outperform quota baskets in welfare. However, as more species are included in the bundle, the per-species welfare gap between instruments narrows, and the total gap settles at a positive constant. As the basket grows, a trade-off emerges between diversifying the uncertainty loss and increasing the mismatch loss relative to a first-best policy. Whether the policymaker uses a tax basket or a quota basket, fishermen keep fishing each period until marginal profits are equalized across all fishable species. This margin integrates biological productivity, harvesting costs, and market prices. These findings help unify a literature that proposes grouping species based on similar ecological (Costello and Szuwalski, 2018), economic (Collado et al., 2021), or vulnerability (Collado et al., 2024) parameters.

In general, both policies are second-best instruments, and tax baskets outperform quota baskets in welfare. Both target a common margin equal to a weighted average of the species' next-period marginal profits, analogous to the optimal uniform tax across multiple pollutants (Diamond, 1973; Fowlie and Muller, 2019; Mendelsohn, 1986). Because the problem is dynamic, the weights can change every period. In this context, a tax basket is particularly helpful because it keeps the remaining biomass (and the weights) constant over time and is immune to ecological shocks. Nevertheless, the tax basket falls further behind species-specific instruments as N increases, because a single tax cannot match many different species-specific optimal taxes. A tax basket achieves the first best only under a specific condition: each species' marginal profit at its own Reed escapement must be the same (equivalently, all individual optimal taxes must coincide). Identical species, or species with very similar economic and biological parameters as Kritzer et al. (2023) recommend, are a special case. By duality, the same condition also makes quota baskets first-best in a deterministic setting.

Quota baskets, by contrast, carry an additional distortion under uncertainty. The timing asymmetry between planner and fisherman does not affect the tax basket because the tax induces the fisherman to stop harvesting when the stock reaches its target escapement, regardless of the shock. However, when the quota basket includes many species and ecological shocks are independent, shocks partly average out, reducing per-species loss relative to the tax basket. In this case, quota baskets approximate the second-best tax on a per-species basis, and the total welfare gap converges to a positive constant as N increases. Diversification is partial when species share an environmental driver: the common part of the shocks does not average out, and the per-species gap converges to a positive floor rather than to zero.

The trade-off between uncertainty diversification and policy mismatch makes the choice of N a central design decision. In the simulations, at empirically calibrated recruitment volatility, the tax outperforms the quota by about 2% of the tax basket's welfare in a two-

species basket but by less than 0.1% at $N = 50$, while the mismatch loss grows to about 1.9% of first-best welfare at $N = 50$. These magnitudes assume independent shocks. When a common driver correlates the shocks, the quota's loss at $N = 50$ rises to 0.6–2.5% of the tax basket's welfare, so the instrument choice matters even in large baskets. While this paper does not determine the optimal basket size, it shows that composition matters. The planner must decide whether the administrative savings justify the mismatch loss. If monitoring capabilities are limited, a bundle instrument is recommended as long as the total welfare loss does not exceed the regulatory cost savings. A larger basket is not always better.

On the other hand, if institutional and scientific capabilities are well developed, a large basket may not be a priority because monitoring one additional species can be relatively cheap when many species are already monitored. This may explain the stark differences in multispecies management (California, China, Alaska, Chile, and Thailand). When datasets are hard to build, managers struggle to distinguish species, or fishing gear is not selective enough, a basket instrument can be attractive. Basket instruments can complement the super-species models (pooling all species into a single stock) used in Southeast Asia (Kulanujaree et al., 2020). However, a basket cannot diversify system-wide shocks. When a common driver, such as El Niño, moves all stocks in the basket together (Fulton et al., 2022), the shocks do not cancel, and the tax basket's per-species advantage no longer vanishes with N (Section 3.3.3). Climate exposure therefore strengthens the case for a tax basket.

These findings are particularly relevant for fisheries characterized by low selectivity, limited data, and high landing aggregation. For example, China harvests many species jointly, and regulators have limited ability to identify or regulate individual stocks (Szuwalski et al., 2017). China's summer fishing moratorium, a seasonal closure applied to essentially all vessels and gear across the Bohai, Yellow, East, and South China Seas (Szuwalski et al., 2017), functions as a whole-fishery effort control, the most aggregated basket possible. Although the model studies landing fees and quotas rather than effort controls, its logic suggests that such an instrument smooths ecological variability across stocks while incurring a mismatch loss that rises with the number of species it sweeps into. This reasoning aligns with broader evidence that diversification across multiple stocks can reduce income volatility and targeting risk for fishers (Cline et al., 2017; Robinson et al., 2020). In contrast, aggregate production models have proven tractable in fisheries such as Thailand's, but doubts remain about their resilience under ecological uncertainty (Fulton et al., 2022). Conceptually, the result parallels findings in environmental economics showing that uniform instruments can perform well when uncertainty is diversified across regulated sources (Fowlie and Muller, 2019).

Few countries employ landing fees, and where they exist, they typically complement rather than replace quota systems (Ben-Hasan et al., 2021), despite Weitzman's (2002) case for prices over quantities. This paper does not overturn that ranking: under ecological uncertainty, the tax basket remains the welfare-preferred instrument. Yet if tax baskets are

politically infeasible, this paper shows when a quota basket comes close to the tax basket per species: in large, balanced baskets of species with independent shocks.

For the policymaker, basket instruments are most attractive in data-limited settings where species-specific regulation is infeasible. Tax baskets do not require real-time stock information at the start of period t or during it. While they reduce monitoring and species-identification costs, harvest-cost parameters are not readily available for all fisheries and may be difficult for authorities to obtain. By duality, quota baskets benefit from aggregation but require periodic updates and reliable aggregate biomass information. This weakens the presumption that basket regulation reduces monitoring costs. Indicator species may partially alleviate this burden, but they introduce additional measurement error that compounds ecological uncertainty and further degrades quota performance.[5] These distinctions suggest that the choice between instruments should depend not only on ecological uncertainty but also on the regulator's institutional and budgetary capacity.

Several limitations point to directions for future research. First, the analysis abstracts from fishing technology choice and gear selectivity. Incorporating endogenous targeting decisions could clarify when basket instruments induce efficiency losses or gains through technological adjustment. Second, the framework focuses on a single basket; extending it to multiple, overlapping baskets raises questions about species mobility across regulatory groups, enforcement, and permit trade and pricing. Third, the model does not represent species interactions. These differ from the correlated shocks in Section 3.3.3. A common environmental driver only changes the covariance of the shocks: each stock still grows from its own escapement, the planner's problem and τ^+ remain unchanged, and only diversification weakens. Predation or competition instead makes growth depend on other stocks, $G^i(y^i, y^{-i})$. This alters the deterministic problem: the Euler equation gains cross-species terms, one species' escapement has value for another, and a single basket price cannot internalize these externalities. Interactions also generate endogenous co-movement, with a lag and possibly a negative sign between competitors. Preliminary derivations suggest that adding them makes the instrument ranking ambiguous. Fourth, linking the marginal-profit condition to trait-based or size-spectrum ecosystem models (Andersen et al., 2016) could help translate the theoretical criteria into empirically observable metrics, informing practical basket design that balances efficiency, biodiversity, and risk. Fifth, some results rest on the form of the cost function, $C^i(s^i) = \frac{\theta^i}{s^i}$, which makes the tax basket stock-independent. Relaxing it to non-constant catchability would likely narrow the tax's advantage. Sixth, both basket instruments require each species' harvesting costs, which, as noted above, planners rarely observe. This is a major implementation obstacle. I am addressing it in ongoing work along two routes. The first evaluates baskets without direct cost data, using econometric methods: because the fisherman equalizes marginal profits across species (Lemma 1), observed prices and catch composition carry information about relative costs. The second

[5] Sethi et al. (2005) demonstrated that measurement uncertainty has the greatest impact on fishery policy, profits, and fishery collapse under single-species management.

develops the theory of tradable quota baskets, in which the price of a basket permit reveals the common margin λ_t (Appendix E.2), information the planner would otherwise have to infer from cost data.

Finally, this work does not consider bycatch of non-valuable or overfished valuable species (e.g., canary and yelloweye rockfish on the U.S. West Coast). For non-valuable species, the equimarginal rule weakens because fishermen can no longer shift catch freely across species. As a result, uncertainty loss would likely fall with lower N. For bycatch of an overfished valuable species, a bundle instrument could delay the recovery of vulnerable species, but it would not strongly restrict extraction from healthier stocks. With differentiated costs, the fisherman decides what to catch, and the regularity condition fails if a stock becomes scarce after a shock. If species enter and exit the fishable set, the outcome depends on the instrument. The simulations show that a species' harvest drops to zero when it is no longer profitable after a negative shock; in that case, the fisherman can compensate by catching other species. In a tax basket, a tax set to include a non-fishable species distorts the margins of the other species. Ideally, the tax would be less stringent and target only the participating species. Thus, the planner must invest in information on participating species, because ecological shocks may change species participation after the policy is set. This creates a trade-off between conservation (designing the policy to include all species) and profit maximization (designing the policy to include only potentially fishable species). In short, bundle instruments cannot fully resolve the sustainability–profitability trade-off they were created to address.

In summary, tax baskets are generally more robust under uncertainty, and both instruments can perform well in highly aggregated systems or under restrictive conditions. However, managing larger baskets involves a trade-off among diversifying uncertainty loss, increasing mismatch loss, and reducing administrative costs. Bundling does not replace protection for vulnerable, non-target stocks. Still, with careful attention to information requirements and ecological risk, basket regulation offers a pragmatic complement to species-specific management. Finally, implementing a basket instrument can be a stepping stone toward more stringent regulation.

References

Abbott, J. K., & Wilen, J. E. (2009). Regulation of fisheries bycatch with common-pool output quotas. Journal of Environmental Economics and Management, 57(2), 195–204. https://doi.org/10.1016/j.jeem.2008.04.003

Ambec, S., & Coria, J. (2013). Prices vs quantities with multiple pollutants. Journal of Environmental Economics and Management, 66(1), 123–140. https://doi.org/10.1016/j.jeem.2012.11.002

Andersen, K. H., Jacobsen, N. S., & Farnsworth, K. (2016). The theoretical foundations for size spectrum models of fish communities [Publisher: NRC Research Press]. Can. J. Fish. Aquat. Sci., 73(4), 575–588. https://doi.org/10.1139/cjfas-2015-0230

Ben-Hasan, A., De La Puente, S., Flores, D., Melnychuk, M. C., Tivoli, E., Christensen, V., Cui, W., & Walters, C. J. (2021). Constrained public benefits from global catch share fisheries [Publisher: Proceedings of the National Academy of Sciences]. Proceedings of the National Academy of Sciences, 118(39), e2021580118. https://doi.org/10.1073/pnas.2021580118

Bonzon, K., McIlwain, K., Strauss, C. K., & Leuvan, T. V. (2013). A guide for managers and fishermen. 1, 188.

Branch, T. A. (2009). How do individual transferable quotas affect marine ecosystems? [_eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1467-2979.2008.00294.x]. Fish and Fisheries, 10(1), 39–57. https://doi.org/10.1111/j.1467-2979.2008.00294.x

Cao, L., Chen, Y., Dong, S., Hanson, A., Huang, B., Leadbitter, D., Little, D. C., Pikitch, E. K., Qiu, Y., Sadovy de Mitcheson, Y., Sumaila, U. R., Williams, M., Xue, G., Ye, Y., Zhang, W., Zhou, Y., Zhuang, P., & Naylor, R. L. (2017). Opportunity for marine fisheries reform in china [Publisher: Proceedings of the National Academy of Sciences]. Proceedings of the National Academy of Sciences, 114(3), 435–442. https://doi.org/10.1073/pnas.1616583114

Christensen, V., & Pauly, D. (2004). Placing fisheries in their ecosystem context, an introduction. Ecological Modelling, 172(2), 103–107. https://doi.org/10.1016/j.ecolmodel.2003.09.002

Clark, C. W. (1990). Mathematical bioeconomics: The optimal management of renewable resources (2nd). Wiley-Interscience.

Clark, C. W., & Kirkwood, G. P. (1986). On uncertain renewable resource stocks: Optimal harvest policies and the value of stock surveys. Journal of Environmental Economics and Management, 13(3), 235–244. https://doi.org/10.1016/0095-0696(86)90033-9

Cline, T. J., Schindler, D. E., & Hilborn, R. (2017). Fisheries portfolio diversification and turnover buffer alaskan fishing communities from abrupt resource and market changes. Nature Communications, 8, 14042. https://doi.org/10.1038/ncomms14042

Collado, M., de la Rosa, G., Leigh, K., & Qiu, S. (2021). Quota baskets: Exploring alternative groupings for fisheries management. Bren Group Project. Retrieved September 15, 2022,

from https://bren.ucsb.edu/projects/quota-baskets-exploring-alternative-groupings-fisheries-management

Collado, M., Free, C., O'Reilly, S., & Gaines, S. (2024). Evaluating the conservation and economic performance of fisheries quota baskets: A belizean fisheries case study. Bren Group Project.

Copes, P. (1986). A critical review of the individual quota as a device in fisheries management [Publisher: [Board of Regents of the University of Wisconsin System, University of Wisconsin Press]]. Land Economics, 62(3), 278–291. https://doi.org/10.2307/3146392

Costello, C., Ovando, D., Clavelle, T., Strauss, C. K., Hilborn, R., Melnychuk, M. C., Branch, T. A., Gaines, S. D., Szuwalski, C. S., Cabral, R. B., Rader, D. N., & Leland, A. (2016). Global fishery prospects under contrasting management regimes [Publisher: Proceedings of the National Academy of Sciences]. Proceedings of the National Academy of Sciences, 113(18), 5125–5129. https://doi.org/10.1073/pnas.1520420113

Costello, C., & Szuwalski, C. (2018). Multispecies management using quota "baskets" [Presentation, Fujian, China].

Diamond, P. A. (1973). Consumption externalities and imperfect corrective pricing. Bell Journal of Economics, 4(2), 526–538.

Edwards, S. F., Link, J. S., & Rountree, B. P. (2004). Portfolio management of wild fish stocks. Ecological Economics, 49(3), 317–329.

FAO. (2025). Review of the state of world marine fishery resources [FAO Fisheries and Aquaculture Technical Paper. Available as PDF]. Food; Agriculture Organization of the United Nations.

Fogarty, M. J., Overholtz, W. J., & Link, J. S. (2012). Aggregate surplus production models for demersal fishery resources of the gulf of maine. Mar. Ecol. Prog. Ser., 459, 247–258. https://doi.org/10.3354/meps09789

Fowlie, M., & Muller, N. (2019). Market-based emissions regulation when damages vary across sources: What are the gains from differentiation? Journal of the Association of Environmental and Resource Economists, 6(3), 607–643. https://doi.org/10.1086/702852

Fulton, E. A., Sainsbury, K., Noranarttragoon, P., Leadbitter, D., Staples, D. J., Porobic, J., Ye, Y., Phoonsawat, R., & Kulanujaree, N. (2022). Shifting baselines and deciding on the desirable form of multispecies maximum sustainable yield. ICES Journal of Marine Science, 79(7), 2138–2154. https://doi.org/10.1093/icesjms/fsac150

Garcia, D., Dolder, P. J., Iriondo, A., Moore, C., Prellezo, R., & Urtizberea, A. (2020). A multi-stock harvest control rule based on "pretty good yield" ranges to support mixed-fisheries management. ICES J. Mar. Sci., 77(1), 119–135. https://doi.org/10.1093/icesjms/fsz181

Gordon, H. S. (1954). The economic theory of a common-property resource: The fishery. Journal of Political Economy, 62(2), 124–142.

Hannesson, R., & Kennedy, J. (2003). Landings fees vs fish quotas [Manuscript, Norwegian Business School, Bergen, Norway].

Hatcher, A. (2022). A model of quota prices in a multispecies fishery with "choke" species and discarding. Environmental and Resource Economics, 82(4), 825–846. https://doi.org/10.1007/s10640-022-00689-8

Hilborn, R., Amoroso, R. O., Anderson, C. M., Baum, J. K., Branch, T. A., Costello, C., de Moor, C. L., Faraj, A., Hively, D., Jensen, O. P., Kurota, H., Little, L. R., Mace, P., McClanahan, T., Melnychuk, M. C., Minto, C., Osio, G. C., Parma, A. M., Pons, M., ... Ye, Y. (2020). Effective fisheries management instrumental in improving fish stock status [Publisher: Proceedings of the National Academy of Sciences]. Proceedings of the National Academy of Sciences, 117(4), 2218–2224. https://doi.org/10.1073/pnas.1909726116

Hilborn, R., Punt, A. E., & Orensanz, J. (2004). Beyond band-aids in fisheries management: Fixing world fisheries. Bull. Mar. Sci., 74(3), 493–507. https://www.ingentaconnect.com/content/umrsmas/bullmar/2004/00000074/00000003/art00003

Hoel, M., & Karp, L. (2002). Taxes versus quotas for a stock pollutant. Resource and Energy Economics, 24(4), 367–384. https://doi.org/10.1016/S0928-7655(02)00014-3

Holzer, J., & DePiper, G. (2019). Intertemporal quota arbitrage in multispecies fisheries. Journal of Environmental Economics and Management, 93, 185–207. https://doi.org/10.1016/j.jeem.2018.12.002

Hoshino, E., van Putten, I., Pascoe, S., & Vieira, S. (2020). Individual transferable quotas in achieving multiple objectives of fisheries management. Marine Policy, 113, 103744. https://doi.org/10.1016/j.marpol.2019.103744

Jin, D., DePiper, G., & Hoagland, P. (2016). Applying portfolio management to implement ecosystem-based fishery management (EBFM). North American Journal of Fisheries Management, 36(3), 652–669. https://doi.org/10.1080/02755947.2016.1146180

Jin, X., Pitcher, T. J., & Cheung, W. W. L. (2020). Fisheries management in china: Review of problems and options. Ocean & Coastal Management, 190, 105157.

Karr, K. A., Miller, V., Coronado, E., Olivares-Bañuelos, N. C., Rosales, M., Naretto, J., Hiriart-Bertrand, L., Vargas-Fernández, C., Alzugaray, R., Puga, R., Valle, S., Osman, L. P., Solís, J. C., Mayorga, M. I., Rader, D., & Fujita, R. (2021). Identifying pathways for climate-resilient multispecies fisheries. Frontiers in Marine Science, 8. Retrieved October 16, 2023, from https://www.frontiersin.org/articles/10.3389/fmars.2021.721883

Kritzer, J. P., Tang, Y., Chen, Y., Costello, C., Gaichas, S., Nies, T., Peñas, E., Sainsbury, K., Shen, C., Szuwalski, C., & Zhu, W. (2023). Advancing multispecies fishery management in china: Lessons from international experience. Aquaculture and Fisheries, 8(3), 351–362. https://doi.org/10.1016/j.aaf.2021.11.004

Kulanujaree, N., Salin, K. R., Noranarttragoon, P., & Yakupitiyage, A. (2020). The transition from unregulated to regulated fishing in thailand. Sustainability, 12(14), 5841. https://doi.org/10.3390/su12145841

Mangin, T., Costello, C., Anderson, J., Arnason, R., Elliott, M., Gaines, S. D., Hilborn, R., Peterson, E., & Sumaila, R. (2018). Are fishery management upgrades worth the cost? [Publisher: Public Library of Science]. PLOS ONE, 13(9), e0204258. https://doi.org/10.1371/journal.pone.0204258

Markowitz, H. (1952). Portfolio selection. Journal of Finance, 7(1), 77–91.

May, R., & Clark, C. (1979). Management of multispecies fisheries. Science, 205(4403). https://doi.org/0.1126/science.205.4403.267

Mendelsohn, R. (1986). Regulating heterogeneous emissions. Journal of Environmental Economics and Management, 13(4), 301–312.

Nakamura, K. (2015). Multispecies mixed stock fisheries management (Technical Report). Marine Stewardship Council. London.

OECD. (2003). The costs of managing fisheries. Organisation for Economic Co-operation; Development. Retrieved October 16, 2023, from https://www.oecd-ilibrary.org/agriculture-and-food/the-costs-of-managing-fisheries_9789264099777-en

Pacific Fishery Management Council. (2024). Pacific coast groundfish fishery management plan for the california, oregon (tech. rep.). Pacific Fishery Management Council. Portland, Oregon. https://www.pcouncil.org/documents/2022/08/pacific-coast-groundfish-fishery-management-plan.pdf

Pauly, D., Christensen, V., Dalsgaard, J., Froese, R., & Torres, F. (1998). Fishing down marine food webs. Science, 279(5352), 860–863. https://doi.org/10.1126/science.279.5352.860

Pauly, D., & Froese, R. (2012). Comments on fao's state of world fisheries and aquaculture (sofia) 2010. Marine Policy, 36(3), 746–752.

Reed, W. J. (1979). Optimal escapement levels in stochastic and deterministic harvesting models. Journal of Environmental Economics and Management, 6(4), 350–363. https://doi.org/10.1016/0095-0696(79)90016-9

Robinson, J. P. W., Robinson, J., Gerry, C., Govinden, R., Freshwater, C., & Graham, N. A. J. (2020). Diversification insulates fisher catch and revenue in heavily exploited tropical fisheries. Science Advances, 6(8), eaaz0587. https://doi.org/10.1126/sciadv.aaz0587

Sanchirico, J. N., Holland, D., Quigley, K., & Fina, M. (2006). Catch-quota balancing in multispecies individual fishing quotas. Marine Policy, 30(6), 767–785. https://doi.org/10.1016/j.marpol.2006.02.002

Sanchirico, J. N., Smith, M. D., & Lipton, D. W. (2008). An empirical approach to ecosystem-based fishery management. Ecological Economics, 64(3), 586–596. https://doi.org/10.1016/j.ecolecon.2007.04.006

Sethi, G., Costello, C., Fisher, A., Hanemann, M., & Karp, L. (2005). Fishery management under multiple uncertainty. Journal of Environmental Economics and Management, 50(2), 300–318. https://doi.org/10.1016/j.jeem.2004.11.005

Singh, R., & Weninger, Q. (2024). Discretion rather than rules in multiple-species fisheries. Journal of Environmental Economics and Management, 125, 102983. https://doi.org/10.1016/j.jeem.2024.102983

Smith, T., Gibbs, M., & Smith, D. (2009). Fishing for more effective incentives [Publisher: American Association for the Advancement of Science]. Science. https://doi.org/10.1126/science.323.5912.337b

Spence, M. A., Thorpe, R. B., Blackwell, P. G., Scott, F., Southwell, R., & Blanchard, J. L. (2021). Quantifying uncertainty and dynamical changes in multi-species fishing mortality rates, catches and biomass by combining state-space and size-based multi-species models. Fish and Fisheries, 22(4), 667–681.

Squires, D., Campbell, H. F., Cunningham, S., Dewees, C. M., Grafton, R. Q., Herrick, S. F., Kirkley, J. E., Pascoe, S., Shallard, B., & Turris, B. R. (1998). Individual transferable quotas in multispecies fisheries. Marine Policy, 22(2), 135–159. https://doi.org/10.1016/S0308-597X(97)00038-4

Su, S., Tang, Y., Chang, B., Zhu, W., & Chen, Y. (2020). Evolution of marine fisheries management in china from 1949 to 2019: How did china get here and where does china go next? Fish and Fisheries, 21(2), 435–452.

Sun, M., Li, Y., & Chen, Y. (2023). Unveiling unselective fishing in china: A nationwide meta-analysis of multispecies fisheries. Fish and Fisheries, 24(1), 142–158. https://doi.org/10.1111/faf.12715

Sun, M., Wo, J., Ren, Y., & Chen, Y. (2025). Hybrid total allowable catch strategy can sustain productive mixed fisheries and conserve both target and non-target species. Can. J. Fish. Aquat. Sci., 82. https://doi.org/10.1139/cjfas-2024-0274

Szuwalski, C. S., Burgess, M. G., Costello, C., & Gaines, S. D. (2017). High fishery catches through trophic cascades in china. Proc Natl Acad Sci U S A, 114(4), 717–721. https://doi.org/10.1073/pnas.1612722114

Thorson, J. T., Jensen, O. P., & Zipkin, E. F. (2014). How variable is recruitment for exploited marine fishes? a hierarchical model for testing life history theory. Canadian Journal of Fisheries and Aquatic Sciences, 71, 973–983. https://doi.org/10.1139/cjfas-2013-0645

Tinbergen, J. (1952). On the theory of economic policy. North-Holland Publishing.

Ulrich, C., Vermard, Y., Dolder, P. J., Brunel, T., Jardim, E., Holmes, S. J., Kempf, A., Mortensen, L. O., Poos, J.-J., & Rindorf, A. (2017). Achieving maximum sustainable yield in mixed fisheries: A management approach for the north sea demersal fisheries. ICES Journal of Marine Science, 74(2), 566–575. https://doi.org/10.1093/icesjms/fsw126

United Nations Conference on Trade and Development. (2022, June). Towards a climate resilient multispecies finfish man [Available as PDF]. United Nations. https://doi.org/10.18356/9789210013666

Weitzman, M. L. (1998). The noah's ark problem. Econometrica, 66(6), 1279. https://doi.org/10.2307/2999617

Weitzman, M. L. (2002). Landing fees vs harvest quotas with uncertain fish stocks. Journal of Environmental Economics and Management, 43(2), 325–338. https://doi.org/10.1006/jeem.2000.1181

Wo, J., Zhang, C., Ji, Y., Xu, B., Xue, Y., & Ren, Y. (2022). A multispecies tac approach to achieving long-term sustainability in multispecies mixed fisheries. ICES Journal of Marine Science, 79(1), 218–229. https://doi.org/10.1093/icesjms/fsab257

Ye, Y., Cochrane, K., Bianchi, G., & Willmann, R. (2013). Rebuilding global fisheries: The china puzzle. Fish and Fisheries, 14(4), 570–579.

Appendix A: Derivation of the optimal tax basket (one species)

A.1: Fisherman's problem

The myopic fisherman maximizes profits by harvesting fish in period t:

$$\pi_t(x) = \max_{h_t \geq 0} \left\{ \int_{x_t - h_t}^{x_t} (p - \tau_t - C(s))\, ds \right\} \tag{33}$$

where p is the unit price of fish, $C(s)$ is the unit cost of harvesting at fish population s, and τ_t is the landing fee per unit of fish. Taking the first-order condition (FOC) with respect to harvest, I obtain the optimality condition for the current period:

$$p - C\left(x_t - h_t\right) = \tau_t$$

Using the definition of escapement $y_t = x_t - h_t$, I can re-express this condition as:

$$p - C\left(y_t\right) = \tau_t \tag{34}$$

The fisherman therefore makes an escapement decision:

$$\begin{aligned} p - \tau_t - C(y_t) &= 0 \\ C(y_t) &= p - \tau_t \\ y_t &= C^{-1}(p - \tau_t) \end{aligned} \tag{35}$$

A.2: Social planner's problem

The social planner picks a landing fee to induce harvest levels that maximize expected present-discounted profits:

$$\mathbb{E}\Big[\sum_{t=0}^{\infty} \beta^t \int_{X_t - h_t}^{X_t} \pi(s)\, ds\Big] \tag{36}$$

where $\beta \equiv \frac{1}{1+\delta} \in (0,1)$, δ is the discount rate, and the stock at the start of period t is $X_t = Z_t G(y_{t-1})$.

A.3: Landing fee

The planner sets τ_t taking into account the fisherman's decision function, with the previous end-of-period escapement $C^{-1}(p - \tau_{t-1})$ as the relevant state variable:

$$V(Y) = \max_{\tau \geq 0} \mathbb{E}\left[\int_{C^{-1}(p-\tau_t)}^{Z_t G(C^{-1}(p-\tau_{t-1}))} \pi(s)\, ds + \beta V(C^{-1}(p - \tau_t)) \right] \tag{37}$$

The profit reduction from the tax and the tax revenue cancel each other. With this, I take the first-order condition with respect to τ_t:

$$\frac{\partial V(.)}{\partial \tau_t} = \mathbb{E}\left[-\left(p - C(y_t(\tau_t))\right)\frac{d}{d\tau_t}\left(y_t(\tau_t)\right)\right] + \beta\mathbb{E}\left[\frac{d}{d\tau_t}[V_{t+1}(.)]\right] = 0 \tag{38}$$

Then, differentiating with respect to τ_{t-1} and using the Leibniz integral rule:

$$\frac{\partial V_t(.)}{\partial \tau_{t-1}} = \mathbb{E}\left[\left(p - C(Z_t G(y_{t-1}(\tau_{t-1})))\right)\frac{d}{d\tau_{t-1}}\left(Z_t G(y_{t-1}(\tau_{t-1}))\right)\right] \tag{39}$$

By the Benveniste–Scheinkman theorem, I move expression (39) forward by one period:

$$\frac{\partial V_{t+1}(.)}{\partial \tau_t} = \mathbb{E}\left[\left(p - C(Z_{t+1} G(y_t(\tau_t)))\right)\frac{d}{d\tau_t}\left(Z_{t+1} G(y_t(\tau_t))\right)\right]$$

Putting everything together:

$$\begin{aligned}
\mathbb{E}\left[\left(p - C(y_t(\tau_t))\right)\frac{d}{d\tau_t}\left(y_t(\tau_t)\right)\right] &= \beta\mathbb{E}\left[\left(p - C(Z_{t+1}G(y_t(\tau_t)))\right)\frac{d}{d\tau_t}\left(Z_{t+1}G(y_t(\tau_t))\right)\right] \\
\mathbb{E}\left[\left(p - C(C^{-1}(p-\tau_t))\right)\frac{d}{d\tau_t}\left(y_t(\tau_t)\right)\right] &= \beta\mathbb{E}\left[\left(p - C(Z_{t+1}G(y_t(\tau_t)))\right)\frac{d}{d\tau_t}\left(Z_{t+1}G(y_t(\tau_t))\right)\right] \text{ Replace } y_t(\tau_t) \\
\mathbb{E}\left[\left(p - (p-\tau_t)\right)\frac{d}{d\tau_t}\left(y_t(\tau_t)\right)\right] &= \beta\mathbb{E}\left[\left(p - C(Z_{t+1}G(y_t(\tau_t)))\right)\frac{d}{d\tau_t}\left(Z_{t+1}G(y_t(\tau_t))\right)\right] \\
\mathbb{E}\left[\tau_t\frac{d}{d\tau_t}\left(y_t(\tau_t)\right)\right] &= \beta\mathbb{E}\left[\left(p - C(Z_{t+1}G(y_t(\tau_t)))\right)\frac{d}{d\tau_t}\left(Z_{t+1}G(y_t(\tau_t))\right)\right] \\
\tau_t\frac{dy_t}{d\tau_t} &= \beta\mathbb{E}\left[\left(p - C(Z_{t+1}G(y_t(\tau_t)))\right)\left[\frac{dZ_{t+1}}{d\tau_t}G(y_t(\tau_t)) + Z_{t+1}\frac{dG(y_t)}{dy_t}\frac{dy_t}{d\tau_t}\right]\right] \\
\tau_t\frac{dy_t}{d\tau_t} &= \beta\mathbb{E}\left[\left(p - C(Z_{t+1}G(y_t(\tau_t)))\right)\left[Z_{t+1}\frac{dG(y_t)}{dy_t}\frac{dy_t}{d\tau_t}\right]\right] \\
\tau_t\frac{dy_t}{d\tau_t} &= \beta\mathbb{E}\left[\left(p - C(Z_{t+1}G(y_t(\tau_t)))\right)\left[Z_{t+1}\frac{dG(y_t)}{dy_t}\right]\right]\frac{dy_t}{d\tau_t} \\
\tau_t &= \beta\mathbb{E}\left[\left(pZ_{t+1}\frac{dG(y_t)}{dy_t} - C(Z_{t+1}G(y_t(\tau_t)))Z_{t+1}\frac{dG(y_t)}{dy_t}\right)\right] \\
\tau_t &= \beta\mathbb{E}\left[\left(pZ_{t+1}\frac{dG(y_t)}{dy_t}\right)\right] - \beta\mathbb{E}\left[C(Z_{t+1}G(y_t(\tau_t)))Z_{t+1}\frac{dG(y_t)}{dy_t}\right] \\
\tau_t &= \beta p\frac{dG(y_t)}{dy_t} - \beta\mathbb{E}\left[Z_{t+1}C(Z_{t+1}G(y_t))\right]\frac{dG(y_t)}{dy_t}
\end{aligned}$$

In the derivation above, the expression is conditioned on period-t information, and taxes do not influence shocks.[6]

[6] A more stringent assumption is independence between Z, G, and C.

To make the result comparable with Reed (1979), I employ a cost function of the family $C(s) = \frac{\theta}{s^\alpha}$. This function is homogeneous: $C(ms) = \frac{\theta}{(ms)^\alpha} = m^{-\alpha}\frac{\theta}{s^\alpha}$. The expression above can then be written in two ways, depending on the catchability assumption:

$$\begin{aligned}
\tau_t &= \beta p \frac{dG(y_t)}{dy_t} - \beta \mathbb{E}\left[Z_{t+1} C(Z_{t+1} G(y_t))\right] \frac{dG(y_t)}{dy_t} \\
\tau_t &= \beta p \frac{dG(y_t)}{dy_t} - \beta \mathbb{E}\left[Z_{t+1} Z_{t+1}^{-\alpha} C(G(y_t))\right] \frac{dG(y_t)}{dy_t} \\
\tau_t &= \beta p \frac{dG(y_t)}{dy_t} - \beta \mathbb{E}\left[Z_{t+1}^{1-\alpha} C(G(y_t))\right] \frac{dG(y_t)}{dy_t} \\
\tau_t &= \beta \left[p - \mathbb{E}\left[Z_{t+1}^{1-\alpha}\right] C(G(y_t))\right] \frac{dG(y_t)}{dy_t}
\end{aligned}$$

For the term $\mathbb{E}\left[Z_{t+1}^{1-\alpha}\right]$, there are two cases:

- With constant catchability, $\alpha = 1$, $\mathbb{E}\left[Z_{t+1}^{1-1}\right] = 1$, and the tax does not depend on the current stocks: $\tau_t = \beta \left[p - (C \circ G)(y_t)\right] \frac{dG(y_t)}{dy_t}$.
- With decreasing catchability, $0 < \alpha < 1$, Jensen's inequality gives $\mathbb{E}\left[Z_{t+1}^{1-\alpha}\right] \leq \mathbb{E}\left[Z_{t+1}\right]^{1-\alpha} = 1$. Hence, $\tau_t \geq \beta \left[p - (C \circ G)(y_t)\right] \frac{dG(y_t)}{dy_t}$: the deterministic expression is a lower bound (a minimum tax).

The right-hand side is part of the Euler equation for constant escapement proposed by Clark (1976). It is then possible to propose a constant tax based on the constant optimal escapement y^* that satisfies the Euler equation:

$$\begin{aligned}
\tau_t &= \beta \left[p - \mathbb{E}\left[Z_{t+1}^{1-\alpha}\right] C(G(y_t))\right] \frac{dG(y_t)}{dy_t} = (p - C(y_t)) \\
\tau^* &= \beta \left[p - \mathbb{E}\left[Z_{t+1}^{1-\alpha}\right] C(G(y^*))\right] \frac{dG(y^*)}{dy^*} = (p - C(y^*)) \\
\tau^* &= \pi(y^*)
\end{aligned}$$

The optimal tax depends only on the optimal escapement, as Weitzman (2002) shows.

Appendix B: Derivation of the optimal marginal profit $m\pi$

B.1: Fisherman's problem

In each period t, a myopic fisherman maximizes profits by harvesting N stocks $i = 1, 2, \ldots, N$ with independent life cycles, given that the planner uses an instrument that sets a target $m\pi_t$:

$$\begin{aligned} \pi_t = \max_{\{h_t^i\}_{i=1}^N} & \left\{ \sum_{i=1}^{N} \int_{X_t^i - h_t^i}^{X_t^i} \left(p^i - m\pi_t - C^i(s^i) \right) ds^i \right\} \\ \text{s.t. } & h_t^i \geq 0 \ \forall i \end{aligned} \tag{40}$$

where $m\pi_t$ is the target marginal profit, which can be implemented through a landing fee per unit of fish or a shared quota applied to all species. From the FOC with respect to each harvest, I obtain the optimality condition for the current period:

$$p^1 - C^1\left(X_t^1 - h_t^1\right) = p^2 - C^2\left(X_t^2 - h_t^2\right) = \cdots = p^N - C^N\left(X_t^N - h_t^N\right) = m\pi_t$$

Using the definition of escapement $y_t^i = X_t^i - h_t^i$, I can re-express this condition as:

$$p^1 - C^1\left(y_t^1\right) = p^2 - C^2\left(y_t^2\right) = \cdots = p^N - C^N\left(y_t^N\right) = m\pi_t \tag{41}$$

Equation (41) is the equimarginal principle. As in the previous section, the fisherman's escapement decision is:

$$\begin{aligned} p^i - m\pi_t - C^i(y_t^i) &= 0 \\ C^i(y_t^i) &= p^i - m\pi_t \\ y_t^i &= (C^i)^{-1}(p^i - m\pi_t) \end{aligned} \tag{42}$$

For a given margin $m\pi_t$, the fisherman's escapement decision is independent of stock size and, therefore, unaffected by the ecological shocks.

B.2: Social planner's problem

The social planner incorporates the fisherman's escapement decision into her dynamic optimization problem:

$$V(\vec{y}_{t-1}) = \max_{m\pi_t \geq 0} \ \mathbb{E}\left\{ \sum_{i=1}^{N} \int_{Y^i(m\pi_t)}^{Z_t^i G^i(y_{t-1}^i)} \left(p^i - C^i(s^i) \right) ds^i + \beta V(\vec{y}_t) \right\} \tag{43}$$

The target marginal profit depends on the available information. If the planner does not know the current stock status, I assume she operates based on each y_{t-1}^i. Although the

planner cannot observe current ecological shocks, I include them in the dynamic problem to evaluate their impact on the tax basket. I take the FOC with respect to $m\pi_t$ (following the same procedure as in Appendix A) and obtain the following Euler equation:

$$\mathbb{E}\Big[\sum_{i=1}^{N}(p^i - C^i(y_t^i))\, y^i_{m\pi_t}\Big] = \beta\mathbb{E}\Big[\sum_{i=1}^{N}\big(p^i - C^i(Z^i_{t+1}G^i(y_t^i))\big)Z^i_{t+1}G^{i\prime}(y_t^i)\, y^i_{m\pi_t}\Big]$$

Replacing the fisherman's condition $y_t^i = (C^i)^{-1}(p^i - m\pi_t)$:

$$\begin{aligned}
\mathbb{E}\Big[\sum_{i=1}^{N}(p^i - C^i(y_t^i))\, y^i_{m\pi_t}\Big] &= \beta\mathbb{E}\Big[\sum_{i=1}^{N}\big(p^i - C^i(Z^i_{t+1}G^i(y_t^i))\big)Z^i_{t+1}G^{i\prime}(y_t^i)\, y^i_{m\pi_t}\Big] \\
\mathbb{E}\Big[\sum_{i=1}^{N}\big(p^i - C^i((C^i)^{-1}(p^i - m\pi_t))\big)\, y^i_{m\pi_t}\Big] &= \beta\mathbb{E}\Big[\sum_{i=1}^{N}\big(p^i - C^i(Z^i_{t+1}G^i(y_t^i))\big)Z^i_{t+1}G^{i\prime}(y_t^i)\, y^i_{m\pi_t}\Big] \\
\mathbb{E}\Big[\sum_{i=1}^{N}\big(p^i - (p^i - m\pi_t)\big)\, y^i_{m\pi_t}\Big] &= \beta\mathbb{E}\Big[\sum_{i=1}^{N}\big(p^i - C^i(Z^i_{t+1}G^i(y_t^i))\big)Z^i_{t+1}G^{i\prime}(y_t^i)\, y^i_{m\pi_t}\Big] \\
\sum_{i=1}^{N} m\pi_t\, y^i_{m\pi_t} &= \beta\mathbb{E}\Big[\sum_{i=1}^{N}\big(p^i - C^i(Z^i_{t+1}G^i(y_t^i))\big)Z^i_{t+1}G^{i\prime}(y_t^i)\, y^i_{m\pi_t}\Big] \\
m\pi_t\sum_{i=1}^{N} y^i_{m\pi_t} &= \beta\mathbb{E}\Big[\sum_{i=1}^{N}\big(p^i - C^i(Z^i_{t+1}G^i(y_t^i))\big)Z^i_{t+1}G^{i\prime}(y_t^i)\, y^i_{m\pi_t}\Big] \\
m\pi_t^+ &= \beta\frac{1}{\sum_{j=1}^{N} y^j_{m\pi_t}}\mathbb{E}\Big[\sum_{i=1}^{N}\big(p^i - C^i(Z^i_{t+1}G^i(y_t^i))\big)Z^i_{t+1}G^{i\prime}(y_t^i)\, y^i_{m\pi_t}\Big] \\
m\pi_t^+ &= \sum_{i=1}^{N}\mathbb{E}\big[\pi^i_{t+1}\big]\frac{y^i_{m\pi_t}}{\sum_{j=1}^{N} y^j_{m\pi_t}}
\end{aligned} \tag{44}$$

If the cost function is $C^i(s^i) = \frac{\theta^i}{s^i}$, then $y^i_{m\pi_t} = \frac{\theta^i}{(p^i - m\pi_t)^2}$:

$$\begin{aligned}
m\pi_t^+ &= \sum_{i=1}^{N}\mathbb{E}\big[\pi^i_{t+1}\big]\frac{y^i_{m\pi_t}}{\sum_{j=1}^{N} y^j_{m\pi_t}} \\
&= \sum_{i=1}^{N}\mathbb{E}\big[\pi^i_{t+1}\big]\frac{\frac{\theta^i}{(p^i - m\pi_t)^2}}{\sum_{j=1}^{N}\frac{\theta^j}{(p^j - m\pi_t)^2}}
\end{aligned} \tag{45}$$

The fisherman's decision is $p^i - m\pi_t = C^i(y_t^i)$, with $C^i(y_t^i) = \frac{\theta^i}{y_t^i}$:

$$
\begin{aligned}
m\pi_t^+ &= \sum_{i=1}^{N} \mathbb{E}\left[\pi_{t+1}^i\right] \frac{\frac{\theta^i}{(p^i - m\pi_t)^2}}{\sum_{j=1}^{N} \frac{\theta^j}{(p^j - m\pi_t)^2}} \\
&= \sum_{i=1}^{N} \mathbb{E}\left[\pi_{t+1}^i\right] \frac{\frac{\theta^i}{C^i(y_t^i)^2}}{\sum_{j=1}^{N} \frac{\theta^j}{C^j(y_t^j)^2}} \\
&= \sum_{i=1}^{N} \mathbb{E}\left[\pi_{t+1}^i\right] \frac{\frac{\left(Y^i(m\pi^+)\right)^2}{\theta^i}}{\sum_{j=1}^{N} \frac{\left(Y^j(m\pi^+)\right)^2}{\theta^j}} \\
&= \sum_{i=1}^{N} \mathbb{E}\left[\pi_{t+1}^i\right] \rho_t^i
\end{aligned}
\tag{46}
$$

The weight of each species depends on its escapement (remaining biomass) and its cost parameter.

B.3: Tax basket equivalence

Replacing $m\pi_t = \tau_t$, it is straightforward to obtain the same expression as in Appendix B.2:

$$
\tau_t = \beta \frac{1}{\sum_{i=1}^{N} \frac{dy_t^i}{d\tau_t}} \sum_{i=1}^{N} \left[p^i - \mathbb{E}\left[(Z_{t+1}^i)^{1-\alpha}\right] C^i(G^i(y_t^i))\right] \frac{dG^i(y_t^i)}{dy_t^i} \frac{dy_t^i}{d\tau_t}
$$

For the term $\mathbb{E}\left[(Z_{t+1}^i)^{1-\alpha}\right]$, there are two cases:

- With constant catchability, $\alpha = 1$, $\mathbb{E}\left[(Z_{t+1}^i)^0\right] = 1$, so the tax does not depend on the current stocks.
- With decreasing catchability, $0 < \alpha < 1$, Jensen's inequality gives $\mathbb{E}\left[(Z_{t+1}^i)^{1-\alpha}\right] \leq \mathbb{E}\left[Z_{t+1}^i\right]^{1-\alpha} = 1$. Hence, the deterministic expression for τ_t is a lower bound (a minimum tax).

The tax basket is the weighted average of the species' marginal profits in the next period. Assuming $\alpha = 1$ and the cost function $C^i = \frac{\theta^i}{s^i}$ gives:

- $y_t^i = \frac{\theta^i}{p^i - \tau_t}$
- $\frac{dy_t^i}{d\tau_t} = \frac{\theta^i}{(p^i - \tau_t)^2}$

From the fisherman's optimization, $p^i - \tau_t = C^i(y_t^i)$ and $C^i(y_t^i) = \frac{\theta^i}{y_t^i}$. With this information, I can rearrange the tax basket:

$$\tau_t = \beta \frac{1}{\sum_{i=1}^{N} \frac{dy_t^i}{d\tau_t}} \sum_{i=1}^{N} \left[p^i - \mathbb{E}\left[(Z_{t+1}^i)^{1-\alpha} \right] C^i(G^i(y_t^i)) \right] \frac{dG^i(y_t^i)}{dy_t^i} \frac{dy_t^i}{d\tau_t}$$

$$\tau_t = \beta \frac{1}{\sum_{i=1}^{N} \frac{dy_t^i}{d\tau_t}} \sum_{i=1}^{N} \left[p^i - C^i(G^i(y_t^i)) \right] \frac{dG^i(y_t^i)}{dy_t^i} \frac{dy_t^i}{d\tau_t} \qquad \text{Replace } \alpha = 1$$

$$\tau_t = \beta \frac{1}{\sum_{i=1}^{N} \frac{\theta^i}{(p^i - \tau_t)^2}} \sum_{i=1}^{N} \left[p^i - C^i(G^i(y_t^i)) \right] \frac{dG^i(y_t^i)}{dy_t^i} \frac{\theta^i}{(p^i - \tau_t)^2} \qquad \text{Replace } \frac{dy_t^i}{d\tau_t} = \frac{\theta^i}{(p^i - \tau_t)^2}$$

$$\tau_t = \beta \frac{1}{\sum_{i=1}^{N} \frac{\theta^i}{C^i(y_t^i)^2}} \sum_{i=1}^{N} \left[p^i - C^i(G^i(y_t^i)) \right] \frac{dG^i(y_t^i)}{dy_t^i} \frac{\theta^i}{C^i(y_t^i)^2} \qquad \text{Replace } p^i - \tau_t = C^i(y_t^i)$$

$$\tau_t = \beta \frac{1}{\sum_{i=1}^{N} \frac{(y_t^i)^2}{\theta^i}} \sum_{i=1}^{N} \left[p^i - C^i(G^i(y_t^i)) \right] \frac{dG^i(y_t^i)}{dy_t^i} \frac{(y_t^i)^2}{\theta^i} \qquad \text{Replace } C^i(y_t^i) = \frac{\theta^i}{y_t^i}$$

$$\tau_t = \frac{1}{\sum_{i=1}^{N} \frac{(y_t^i(\tau_t))^2}{\theta^i}} \sum_{i=1}^{N} \pi_{t+1}^i(y_t^i(\tau_t)) \frac{(y_t^i(\tau_t))^2}{\theta^i} \qquad \text{Rename next period's marginal profit}$$

$$\tau_t = \tau_t(\vec{y}_t)$$

As with $m\pi_t$, the tax basket is a function of each species' escapement. If the species are identical, the tax basket equals the landing fee for an individual species, as shown in Appendix A. Moreover, the social planner does not need to know the ecological shocks in the current period.

Appendix C: Auxiliary proofs

C.1: Welfare rearrangement

By Lemma 1, the fisherman equates $p^i - C^i(y_t^i) = m\pi_t$, with $m\pi_t = \tau_t$ under the tax and $m\pi_t = \lambda_t$ under the quota, so $y_t^i = Y^i(m\pi_t)$. If $X_t^i > y_t^i$ for all i, the lemma holds every period. Therefore, escapement depends on the ecological shocks only through the common margin. With $C^i(s) = \theta^i/s$, the fisherman's myopic payoff is

$$\int_{y_t^i}^{X_t^i} \Big(p^i - \frac{\theta^i}{s}\Big) ds = A^i(X_t^i) + \phi^i(y_t^i), \quad \text{where } A^i(X) := p^i X - \theta^i \ln X, \quad \phi^i(y) := -p^i y + \theta^i \ln y,$$

which is (15). The current stock and the current escapement enter additively. Substituting $X_t^i = Z_t^i G^i(y_{t-1}^i)$, and using $\mathbb{E}[Z_t^i] = 1$, $\mathbb{E}|\ln Z_t^i| < \infty$, and $\ln(Z_t^i G^i) = \ln Z_t^i + \ln G^i$,

$$\begin{aligned}
\mathbb{E}\big[A^i\big(Z_t^i G^i(y_{t-1}^i)\big)\big] &= \mathbb{E}\big[p^i X - \theta^i \ln X\big] \\
&= \mathbb{E}\big[p^i Z_t^i G^i - \theta^i \ln(Z_t^i G^i)\big] \\
&= \mathbb{E}\big[p^i Z_t^i G^i - \theta^i \ln G^i - \theta^i \ln Z_t^i\big] \\
&= p^i G^i(y_{t-1}^i) - \theta^i \ln G^i(y_{t-1}^i) - \theta^i \,\mathbb{E}[\ln Z_t^i] \\
&= a^i(y_{t-1}^i) - \kappa^i
\end{aligned}$$

with $a^i(y) := p^i G^i(y) - \theta^i \ln G^i(y)$, $\kappa^i := \theta^i \mathbb{E}[\ln Z_t^i]$, and $|\mathbb{E}[\ln Z_t^i]| < \infty$ by assumption. Then, taking the initial stock X_0^i (and y_0^i) as given and summing across species,

$$\mathcal{W} = \sum_i \Big\{ A^i(X_0^i) + \phi^i(y_0^i) + \sum_{t\geq 1} \beta^t \big[a^i(y_{t-1}^i) - \kappa^i + \phi^i(y_t^i)\big] \Big\}$$

Shifting the index in $\sum_{t\geq 1} \beta^t a^i(y_{t-1}^i) = \beta \sum_{t\geq 0} \beta^t a^i(y_t^i)$ and reallocating β^t,

$$\begin{aligned}
\sum_{t\geq 1} \beta^t \big[a^i(y_{t-1}^i) - \kappa^i + \phi^i(y_t^i)\big] &= \underbrace{\sum_{t\geq 1} \beta^t a^i(y_{t-1}^i)}_{\text{(I): stock block}} - \underbrace{\sum_{t\geq 1} \beta^t \kappa^i}_{\text{(II): shock block}} + \underbrace{\sum_{t\geq 1} \beta^t \phi^i(y_t^i)}_{\text{(III): escapement block}} \\
&= \beta \sum_{t\geq 0} \beta^t a^i(y_t^i) - \kappa^i \sum_{t\geq 1} \beta^t + \sum_{t\geq 1} \beta^t \phi^i(y_t^i) \\
&= \beta \sum_{t\geq 0} \beta^t a^i(y_t^i) - \kappa^i \beta \sum_{t\geq 0} \beta^t + \sum_{t\geq 1} \beta^t \phi^i(y_t^i) \\
&= \beta \sum_{t\geq 0} \beta^t a^i(y_t^i) - \kappa^i \frac{\beta}{1-\beta} + \sum_{t\geq 1} \beta^t \phi^i(y_t^i)
\end{aligned}$$

Putting things together:

$$
\begin{aligned}
\mathcal{W} &= \sum_i \Big\{ A^i(X_0^i) + \phi^i(y_0^i) + \beta \sum_{t\geq 0} \beta^t a^i(y_t^i) - \kappa^i \frac{\beta}{1-\beta} + \sum_{t\geq 1} \beta^t \phi^i(y_t^i) \Big\} \\
&= \sum_i A^i(X_0^i) + \sum_i \phi^i(y_0^i) + \sum_i \beta \sum_{t\geq 0} \beta^t a^i(y_t^i) - \sum_i \kappa^i \frac{\beta}{1-\beta} + \sum_i \sum_{t\geq 1} \beta^t \phi^i(y_t^i) \\
&= \sum_i A^i(X_0^i) + \sum_i \phi^i(y_0^i) + \sum_{t\geq 0} \beta^t \sum_i \beta a^i(y_t^i) - \sum_i \kappa^i \frac{\beta}{1-\beta} + \sum_{t\geq 1} \beta^t \sum_i \phi^i(y_t^i) \\
&= \sum_i \Big[A^i(X_0^i) - \kappa^i \frac{\beta}{1-\beta} \Big] + \sum_{t\geq 0} \beta^t \sum_i \beta a^i(y_t^i) + \sum_{t\geq 0} \beta^t \sum_i \phi^i(y_t^i) \\
&= \underbrace{\sum_i \Big[A^i(X_0^i) - \kappa^i \frac{\beta}{1-\beta} \Big]}_{=:\Xi_0} + \sum_{t\geq 0} \beta^t \sum_i \big[\underbrace{\phi^i(y_t^i) + \beta a^i(y_t^i)}_{=:w^i(y_t^i)} \big] \\
&= \Xi_0 + \sum_{t\geq 0} \beta^t \sum_i w^i(y_t^i)
\end{aligned}
\tag{47}
$$

This result applies to both the tax and the quota basket.

C.2: Recovering the dynamic solution through the static problem

Returning to equation (47) and assuming that each $y_t^i = y^i$ is constant over time:

$$
\begin{aligned}
\mathcal{W} &= \sum_i \Big\{ A^i(X_0^i) - \kappa^i \frac{\beta}{1-\beta} + \frac{1}{1-\beta} \Big[\phi^i(y^i) + \beta a^i(y^i) \Big] \Big\} \\
&= \sum_i \Big\{ A^i(X_0^i) - \kappa^i \frac{\beta}{1-\beta} + \frac{1}{1-\beta} \Big[\beta \big(p^i G^i(y^i) - \theta^i \ln G^i(y^i) \big) - \big(p^i y^i - \theta^i \ln y^i \big) \Big] \Big\} \\
&= \sum_i \Big\{ A^i(X_0^i) - \kappa^i \frac{\beta}{1-\beta} + \underbrace{\frac{1}{1-\beta} \Big[\beta p^i G^i(y^i) - \beta \theta^i \ln G^i(y^i) - p^i y^i + \theta^i \ln y^i \Big]}_{\mathcal{V}^i} \Big\}
\end{aligned}
\tag{48}
$$

I take the derivative with respect to each y^i:

$$
\begin{aligned}
\frac{\partial \mathcal{W}}{\partial y^i} &= \frac{1}{1-\beta}\Big[\beta p^i G^{i\prime}(y^i) - \beta\theta^i \frac{1}{G^i(y^i)} G^{i\prime}(y^i) - p^i + \theta^i \frac{1}{y^i}\Big] \\
&= \frac{1}{1-\beta}\Big[\beta\Big(p^i - \theta^i \frac{1}{G^i(y^i)}\Big) G^{i\prime}(y^i) - \Big(p^i - \theta^i \frac{1}{y^i}\Big)\Big] \\
&= \frac{1}{1-\beta}\Big[\beta\big(p^i - C^i(G^i(y^i))\big) G^{i\prime}(y^i) - \big(p^i - C^i(y^i)\big)\Big] \\
&= \frac{1}{1-\beta}\Big[\pi^i_{t+1}(y^i) - \big(p^i - C^i(y^i)\big)\Big]
\end{aligned}
\tag{49}
$$

Setting (49) to zero recovers Reed's condition for species i; imposing a common margin $y^i = Y^i(\tau)$ and weighting by $dY^i/d\tau$ recovers the Euler equation of Proposition 2. For species-specific management, the discount factor multiplies the growth block in (48), which is what makes (49) vanish exactly at Reed's y^{i*}. Because $\pi^{i\prime}_{t+1} \leq 0$ and $C^{i\prime} < 0$, $\mathcal{V}^{i\prime\prime} < 0$: $\mathcal{V}^i$ is strictly concave with unique maximizer y^{i*}. Hence,

$$
\pi^i_{t+1}(y^i) > p^i - C^i(y^i) \;\text{ for } y^i < y^{i*}, \qquad \pi^i_{t+1}(y^i) < p^i - C^i(y^i) \;\text{ for } y^i > y^{i*} \tag{50}
$$

This sign rule is key to proving uniqueness.

Appendix D: Existence, uniqueness and interiority of τ^+

Given Proposition 2 and Appendix C.2, the regularity condition $\max_i \tau^{i*} < \tau^F_{\max} := \min_i(p^i - C^i(K^i))$, and the cost function $C^i = \theta^i/s^i$, I can rewrite the tax optimality condition as a residual function:

$$g(\tau) := \tau - R(\tau), \qquad \text{where } R(\tau) := \frac{\sum_i \pi^i(Y^i(\tau))\,(Y^i(\tau))^2/\theta^i}{\sum_i (Y^i(\tau))^2/\theta^i}.$$

Since $\mathcal{W}'(\tau) = -\big(\sum_i Y^{i\prime}(\tau)\big)g(\tau)$, g and $\mathcal{W}'$ have the same roots and opposite signs.

D.1: Endpoint signs

From Appendix C.2, the tax solution commits species i to a constant escapement y forever. Without loss of generality, order species by their individual optimal taxes, $\tau^{1*} \le \cdots \le \tau^{N*}$, and assume not all τ^{i*} are equal. Then:

1. $Y^i(\cdot)$ is strictly increasing, so $\tau < \tau^{i*} \iff Y^i(\tau) < y^{i*}$ (a lower tax leaves less stock in the ocean).
2. At $\tau = \tau^{1*}$, species 1 contributes exactly $\pi^1_{t+1}(Y^1(\tau^{1*})) = \tau^{1*}$ (shown in Appendix A).
3. Each $j \ge 2$ has $\tau^{1*} < \tau^{j*}$, hence $Y^j(\tau^{1*}) < y^{j*}$ (overharvest), and (50) with $p^j - C^j(Y^j(\tau^{1*})) = \tau^{1*}$ gives $\pi^j_{t+1}(Y^j(\tau^{1*})) > \tau^{1*}$.
4. A weighted average with strictly positive weights (guaranteed by the fishable range) of terms that are all $\ge \tau^{1*}$, at least one strictly, exceeds τ^{1*}. Therefore $R(\tau^{1*}) > \tau^{1*}$ and $g(\tau^{1*}) < 0$.
5. At the opposite endpoint, $\tau = \tau^{N*}$, each $j < N$ symmetrically has $Y^j(\tau^{N*}) > y^{j*}$ (underharvest), so $\pi^j_{t+1}(Y^j(\tau^{N*})) < \tau^{N*}$, giving $R(\tau^{N*}) < \tau^{N*}$ and $g(\tau^{N*}) > 0$.

D.2: Existence

Y^i, π^i, and the denominator $\sum_i \theta^i/(p^i - \tau)^2$ are continuous on $[\tau^{1*}, \tau^{N*}]$. The interior regularity condition guarantees $p^i - \tau \ge C^i(K^i) > 0$; the denominator is then bounded and bounded away from zero, and g is continuous. By the Intermediate Value Theorem, there exists $\tau^+ \in (\tau^{1*}, \tau^{N*})$ with $g(\tau^+) = 0$.

D.3: Uniqueness

A sufficient condition is that $\mathcal{W}$ be strictly concave in τ on $[\tau^{1*}, \tau^{N*}]$. Concavity of each w^i in its own escapement (Appendix C.2) is not enough, because τ enters w^i through Y^i,

and $Y^{i\prime}$ sets the weights. Dropping Ξ_0 and the discounting, which do not depend on τ, I focus on per-period basket welfare, $\mathcal{W}(\tau) = \sum_i w^i(Y^i(\tau))$, defined in (24). Differentiating $\mathcal{W}$ twice:

$$\mathcal{W}''(\tau) = \sum_i \Big[\underbrace{w^{i\prime\prime}(y^i)\,\big(Y^{i\prime}(\tau)\big)^2}_{\leq 0 \text{ if } w^i \text{ concave}} + \underbrace{w^{i\prime}(y^i)\,Y^{i\prime\prime}(\tau)}_{\text{indefinite}} \Big], \qquad y^i = Y^i(\tau) \tag{51}$$

The y^i differ across species even though τ is shared, and the sum contains no cross terms ($\mathcal{W}$ is separable). Per-species concavity is therefore sufficient for global concavity. However, the term $w^{i\prime}(y^i)\,Y^{i\prime\prime}(\tau)$ has an ambiguous sign: $Y^{i\prime\prime}(\tau) = 2\theta^i/(p^i-\tau)^3 > 0$, while $w^{i\prime}(y^i) = \pi^i_{t+1}(y^i) - \tau$ is positive below τ^{i*} and negative above it by (50).

To handle this, I remove the nonlinearity of Y^i through an affine reparametrization. From the fisherman's rule $p^i - \theta^i/y^i = \tau$,

$$p^i - \frac{\theta^i}{y^i} = \tau$$
$$z^i := \frac{1}{y^i} = \frac{p^i - \tau}{\theta^i}$$

is affine in τ, so $z^{i\prime\prime}(\tau) = 0$ and the curvature term vanishes. I rewrite $\mathcal{W}$:

$$\mathcal{W}(\tau) = \sum_i w^i(y^i) = \sum_i w^i(1/z^i) = \sum_i \hat{w}^i(z^i), \quad \text{where } \hat{w}^i(z) := w^i(1/z).$$

Because each z^i is affine in τ and $\mathcal{W}$ is separable, $\mathcal{W}''(\tau) = \sum_i \hat{w}^{i\prime\prime}(z^i)\,(z^{i\prime})^2$ with $(z^{i\prime})^2 > 0$. Hence $\mathcal{W}$ is concave in τ if $\hat{w}^{i\prime\prime}(z^i) \leq 0$ for all i, and strictly concave if, in addition, the inequality is strict for at least one i. Using $\dfrac{dy}{dz} = \dfrac{d}{dz}z^{-1} = -\dfrac{1}{z^2}$ and $z = 1/y \Rightarrow \dfrac{1}{z^2} = y^2$,

$$\hat{w}^{i\prime}(z) = w^{i\prime}\big(\tfrac{1}{z}\big)\,\big(-\tfrac{1}{z^2}\big) = -\tfrac{1}{z^2}\,w^{i\prime}(y)$$

and

$$\hat{w}^{i\prime\prime}(z) = \tfrac{2}{z^3}w^{i\prime}\big(\tfrac{1}{z}\big) - \tfrac{1}{z^2}w^{i\prime\prime}\big(\tfrac{1}{z}\big)\,\big(-\tfrac{1}{z^2}\big) = \tfrac{2}{z^3}w^{i\prime}\big(\tfrac{1}{z}\big) + \tfrac{1}{z^4}w^{i\prime\prime}\big(\tfrac{1}{z}\big)\,.$$

Replacing $z = 1/y$ (so $1/z^3 = y^3$, $1/z^4 = y^4$),

$$\begin{aligned}\hat{w}^{i\prime\prime}(z) &= 2y^3 w^{i\prime}(y) + y^4 w^{i\prime\prime}(y) \\ &= y^3\big[\,y\,w^{i\prime\prime}(y) + 2w^{i\prime}(y)\,\big]\end{aligned}$$

From Appendix C.2, $w^{i\prime}(y) = \pi^i_{t+1}(y) - p^i + \theta^i/y$ and $w^{i\prime\prime}(y) = \pi^{i\prime}_{t+1}(y) - \theta^i/y^2$, so

$$\begin{aligned}\hat{w}^{i\prime\prime}(z) &= y^3\left[\, y\,(\pi^{i\prime}_{t+1}(y) - \theta^i/y^2) + 2(\pi^i_{t+1}(y) - p^i + \theta^i/y)\right] \\ &= y^3\left[y\pi^{i\prime}_{t+1}(y) - \theta^i/y + 2\pi^i_{t+1}(y) - 2p^i + 2\theta^i/y\right] \\ &= y^3\left[y\pi^{i\prime}_{t+1}(y) + \theta^i/y + 2\pi^i_{t+1}(y) - 2p^i\right] \\ &= y^3\left[y\pi^{i\prime}_{t+1}(y) + C^i(y) + 2\pi^i_{t+1}(y) - 2p^i\right].\end{aligned}$$

Using the fisherman's species-specific optimality condition $C^i(Y^i(\tau)) = p^i - \tau$,

$$\begin{aligned}\hat{w}^{i\prime\prime}(z) &= y^3\left[y\pi^{i\prime}_{t+1}(y) + p^i - \tau + 2\pi^i_{t+1}(y) - 2p^i\right] \\ &= \underbrace{y^3}_{>0}\,\underbrace{\left[y\pi^{i\prime}_{t+1}(y) + 2\pi^i_{t+1}(y) - p^i - \tau\right]}_{\text{sign of } \hat{w}^{i\prime\prime}}.\end{aligned}$$

Since $y^3 > 0$, $\hat{w}^{i\prime\prime}(z) \le 0$ if and only if the bracket is ≤ 0, which gives:

Condition for global concavity. For every i and every $\tau \in [\tau^{1*}, \tau^{N*}]$, writing $y = Y^i(\tau)$,

$$y\,\pi^{i\prime}_{t+1}(y) + 2\pi^i_{t+1}(y) \;\le\; p^i + \tau, \tag{52}$$

with strict inequality for at least one i.

Because $\mathcal{W}$ is separable, imposing (52) for every i makes $\mathcal{W}''(\tau) < 0$.

D.4: Interiority

The endpoint signs $g(\tau^{1*}) < 0 < g(\tau^{N*})$ are strict, so $\tau^+ \in (\tau^{1*}, \tau^{N*}) \subset (0, \tau^F_{\max})$. The optimum is therefore never at $\min_i \tau^{i*}$ or $\max_i \tau^{i*}$ unless these coincide.

Appendix E: Derivation of the optimal quota basket

E.1: Fisherman's problem

Each period, the fisherman maximizes profits over the harvest of N stocks subject to the shared quota:

$$\begin{aligned} \max_{\{h_t^i\}} \left\{ \pi_t = \sum_i \Big(p^i h_t^i - \int_{y_t^i}^{X_t^i} C^i(s^i)\, ds^i \Big) \right\} \\ \text{subject to} \quad \omega_t = \sum_i h_t^i = \sum_i X_t^i - \sum_i y_t^i \end{aligned} \tag{53}$$

The Lagrangian is

$$\mathcal{L} = \sum_i \Big(p^i h_t^i - \int_{y_t^i}^{X_t^i} C^i(s^i)\, ds^i \Big) + \lambda_t \Big(\omega_t - \sum_i h_t^i \Big),$$

with FOC for each stock

$$\begin{aligned} p^i - C^i(y_t^i) &= \lambda_t \\ y_t^i &= (C^i)^{-1}(p^i - \lambda_t) = Y^i(\lambda_t) \end{aligned} \tag{54}$$

The realized stocks $X_t^i = Z_t^i G^i(y_{t-1}^i)$ depend on last period's escapement and this period's shock, not on the current quota:

$$\begin{aligned} \omega_t &= \sum_i X_t^i - \sum_i Y^i(\lambda_t) \\ \sum_i Y^i(\lambda_t) &= \sum_i X_t^i - \omega_t \\ \lambda_t &= \lambda(\vec{X}_t, \omega_t) \end{aligned} \tag{55}$$

Differentiating (55) with respect to ω_t, holding the stocks $\vec{X}_t$ fixed (they are predetermined when the fisherman harvests),

$$\begin{aligned} \sum_i Y^i(\lambda_t) &= \sum_i X_t^i - \omega_t \\ \frac{\partial}{\partial \omega_t} \Big(\sum_i Y^i(\lambda_t) \Big) &= \frac{\partial}{\partial \omega_t} \Big(\sum_i X_t^i - \omega_t \Big) \\ \Big(\sum_i Y_\lambda^i \Big) \lambda_{\omega_t} &= -1 \\ \lambda_{\omega_t} &= \frac{-1}{\sum_i Y_\lambda^i} < 0 \end{aligned} \tag{56}$$

If the cost function is $C^i = \theta^i/s^i$, $Y^i_\lambda = \theta^i/(p^i - \lambda)^2 = (y^i)^2/\theta^i > 0$, because $p^i - \lambda = C^i(y^i) = \theta^i/y^i$ and $y^i, \theta^i > 0$. Consequently, a larger quota lowers the common shadow value. A change in the harvest limit causes the following change in each species' escapement:

$$\begin{aligned} y^i_t &= Y^i(\lambda_t) \\ \frac{\partial y^i_t}{\partial \omega_t} &= Y^i_\lambda \, \lambda_{\omega_t} \\ &= Y^i_\lambda \; \frac{-1}{\sum_j Y^j_\lambda} && \text{Replace (56)} \\ &= -\frac{(y^i_t)^2/\theta^i}{\sum_j (y^j_t)^2/\theta^j} \\ &= -\rho^i_t && \text{Rename} \end{aligned} \tag{57}$$

By construction, $\sum_i \rho^i_t = 1$; the regularity condition ensures that every species enters the sum. The weights ρ^i_t record how the fisherman spreads one extra unit of the quota basket across species.

E.2: Choosing the quota is choosing the shared shadow value

Total profits per period are:

$$\Pi(\vec{X}_t, \omega_t) := \sum_i \int_{y^i_t}^{X^i_t} \left(p^i - C^i(s)\right) ds$$

with $y^i_t = Y^i(\lambda_t)$. Differentiating with respect to ω_t, and using the fisherman's FOC $p^i - C^i(y^i_t) = \lambda_t$ and $\sum_i \partial y^i_t/\partial \omega_t = -1$ from (55),

$$\begin{aligned} \frac{\partial \Pi}{\partial \omega_t} &= -\sum_i \left(p^i - C^i(y^i_t)\right) \frac{\partial y^i_t}{\partial \omega_t} \\ &= -\sum_i \lambda_t \frac{\partial y^i_t}{\partial \omega_t} \\ &= -\lambda_t \sum_i \frac{\partial y^i_t}{\partial \omega_t} \\ &= -\lambda_t \sum_i (-\rho^i_t) \\ &= \lambda_t \end{aligned} \tag{58}$$

The marginal social value of an extra unit of aggregate quota equals the fisherman's shadow value λ_t. This is the duality between the two derivations: λ_t is the price that decentralizes the quota.

E.3: Social planner's problem

Take the predetermined escapement vector $\vec{y}_{t-1}$ as the state. The planner sets the quota ω_t before the shocks, so ω_t is deterministic while λ_t and y_t^i are random through $\vec{X}_t$:

$$V(\vec{y}_{t-1}) = \max_{\omega_t \geq 0} \mathbb{E}\Big[\Pi(\vec{X}_t, \omega_t) + \beta V(\vec{y}_t)\Big], \qquad X_t^i = Z_t^i G^i(y_{t-1}^i), \quad y_t^i = Y^i\big(\lambda(\vec{X}_t, \omega_t)\big) \tag{59}$$

Only y_t^i depends on ω_t (the stocks $\vec{X}_t$ are predetermined by $\vec{y}_{t-1}$). Using $\partial\Pi/\partial\omega_t = \lambda_t$ from (58) and $\partial y_t^i/\partial\omega_t = -\rho_t^i$ from (57), I derive the FOC with respect to the quota basket:

$$\begin{aligned} \frac{\partial V(\cdot)}{\partial \omega_t} &= \mathbb{E}\Big[\frac{\partial \Pi}{\partial \omega_t} + \beta[V_{y^1} y_\lambda^1 \lambda_\omega + V_{y^2} y_\lambda^2 \lambda_\omega + \cdots + V_{y^N} y_\lambda^N \lambda_\omega]\Big] = 0 \\ &= \mathbb{E}\Big[\frac{\partial \Pi}{\partial \omega_t} + \beta \sum_j V_{y^j}(\vec{y}_t)\, \frac{\partial y_t^j}{\partial \omega_t}\Big] = 0 \\ &= \mathbb{E}\Big[\lambda_t - \beta \sum_j \rho_t^j\, V_{y^j}(\vec{y}_t)\Big] = 0 \end{aligned} \tag{60}$$

The optimal quota basket equates, in expectation, the current shadow value of the quota with the ρ-weighted continuation value of the escapement.

The state y_{t-1}^k enters (59) through $X_t^k = Z_t^k G^k(y_{t-1}^k)$. Differentiating (55) with respect to the stock, holding ω_t fixed,

$$\Big(\sum_j Y_\lambda^j\Big)\frac{\partial \lambda_t}{\partial X_t^k} = 1, \qquad \frac{\partial y_t^j}{\partial X_t^k} = Y_\lambda^j \frac{\partial \lambda_t}{\partial X_t^k} = \frac{Y_\lambda^j}{\sum_m Y_\lambda^m} = \rho_t^j, \qquad \sum_j \frac{\partial y_t^j}{\partial X_t^k} = 1 \tag{61}$$

Applying the envelope theorem (the indirect effect through the optimal ω_t is zero by (60)) and the chain rule,

$$
\begin{aligned}
V_{y^k_{t-1}}(\vec{y}_{t-1}) &= \mathbb{E}\Big[\Big(\frac{\partial \Pi}{\partial X^k_t} + \beta \frac{\partial V(\vec{y}_t)}{\partial X^k_t}\Big) \underbrace{Z^k_t G^{k\prime}(y^k_{t-1})}_{\partial X^k_t / \partial y^k_{t-1}}\Big] \\
&= \mathbb{E}\Big[\Big(\sum_i \big(p^i - C^i(X^i_t)\big)\frac{\partial X^i_t}{\partial X^k_t} - \sum_j \big(p^j - C^j(y^j_t)\big)\frac{\partial y^j_t}{\partial \lambda_t}\frac{\partial \lambda_t}{\partial X^k_t} \;+\; \beta \sum_j V_{y^j}(\vec{y}_t)\frac{\partial y^j_t}{\partial X^k_t}\Big) Z^k_t G^{k\prime}(y^k_{t-1})\Big] \\
&= \mathbb{E}\Big[\Big(\sum_i \big(p^i - C^i(X^i_t)\big)\delta_{ik} - \sum_j \big(p^j - C^j(y^j_t)\big)\frac{\partial y^j_t}{\partial X^k_t} \;+\; \beta \sum_j V_{y^j}(\vec{y}_t)\frac{\partial y^j_t}{\partial X^k_t}\Big) Z^k_t G^{k\prime}(y^k_{t-1})\Big] \\
&= \mathbb{E}\Big[\Big(p^k - C^k(X^k_t) - \sum_j \big(p^j - C^j(y^j_t)\big)\frac{\partial y^j_t}{\partial X^k_t} \;+\; \beta \sum_j V_{y^j}(\vec{y}_t)\frac{\partial y^j_t}{\partial X^k_t}\Big) Z^k_t G^{k\prime}(y^k_{t-1})\Big] \\
&= \mathbb{E}\Big[\Big(p^k - C^k(X^k_t) - \lambda_t \sum_j \rho^j_t + \beta \sum_j \rho^j_t V_{y^j}(\vec{y}_t)\Big) Z^k_t G^{k\prime}(y^k_{t-1})\Big] \\
&= \mathbb{E}\Big[\Big(p^k - C^k(X^k_t) - \lambda_t + \beta \sum_j \rho^j_t V_{y^j}(\vec{y}_t)\Big) Z^k_t G^{k\prime}(y^k_{t-1})\Big]
\end{aligned}
\tag{62}
$$

using $\partial X^i_t / \partial X^k_t = \delta_{ik}$ for the stocks (only stock k responds to its own y^k_{t-1}) and $\partial y^j_t / \partial X^k_t = \rho^j_t$ from (61) for the quota-coupled escapements (all respond, through λ_t). The bracket is the value of one extra fish in stock k, minus the shadow value λ_t of the escapement it displaces, plus its continuation value. The escapements are all tied together through the quota basket.

Equations (60)–(62) are the direct-ω analog of Appendix B.3.

E.4: Recovering λ^+

With $Z^i_t \equiv 1$, the expectation drops and (60) holds pointwise, $\lambda_t = \beta \sum_j \rho^j_t V_{y^j}(\vec{y}_t)$. Substituting this into (62) cancels the last two bracketed terms, leaving

$$
\begin{aligned}
V_{y^i_{t-1}}(\vec{y}_{t-1}) &= \mathbb{E}_{Z_t}\Big[\Big(p^i - C^i(X^i_t) - \lambda_t + \beta \sum_j \rho^j_t\, V_{y^j}(\vec{y}_t)\Big) Z^i_t G^{i\prime}(y^i_{t-1})\Big] \\
&= \big(p^i - C^i(X^i_t) - \lambda_t + \lambda_t\big) G^{i\prime}(y^i_{t-1}) \\
&= \big(p^i - C^i(G^i(y^i_{t-1}))\big) G^{i\prime}(y^i_{t-1}) \\
&= \frac{1}{\beta}\pi^i_{t+1}(y^i_{t-1}) \quad \text{using the definition } \pi^i_{t+1} = \beta\big(p^i - C^i(G^i)\big)G^{i\prime} \text{ with } Z \equiv 1
\end{aligned}
\tag{63}
$$

Substituting back into the FOC and using Remark 1.2,

$$
\begin{aligned}
\lambda_t &= \beta \sum_i \rho_t^i V_{y^i}(\vec{y}_t) \\
\lambda_t &= \sum_i \rho_t^i \, \pi_{t+1}^i(y_t^i) \\
\lambda^+ &= \frac{\sum_i \pi_{t+1}^i(y_t^i)\,(y_t^i)^2/\theta^i}{\sum_j (y_t^j)^2/\theta^j}
\end{aligned}
\tag{64}
$$

which is exactly the planner's FOC for the tax, (14). Like the optimal tax basket, the optimal shadow value is fixed over time, $\lambda_t = \lambda^+$, with $y_t^i = Y^i(\lambda^+)$, if the regularity condition holds.

E.5: The quota basket is second-best

Under uncertainty, the collapse in Appendix E.4 fails because the shadow value λ_t^F and the reallocation weights ρ_t^i are both functions of the realized shocks. The planner, by contrast, commits to a deterministic ω_t. The FOC residual is

$$
R_t := \lambda_t^F - \beta \sum_j \rho_t^j V_{y^j}(\vec{y}_t),
$$

so that (60) reads $\mathbb{E}[R_t] = 0$. Substituting into (62) and separating,

$$
\begin{aligned}
V_{y^k}(\vec{y}_{t-1}) &= \mathbb{E}\Big[\Big(p^k - C^k(X_t^k) - \lambda_t + \beta \sum_j \rho_t^j V_{y^j}(\vec{y}_t)\Big) Z_t^k G^{k\prime}(y_{t-1}^k)\Big] \\
&= \mathbb{E}\Big[\Big(p^k - C^k(X_t^k) - R_t\Big) Z_t^k G^{k\prime}(y_{t-1}^k)\Big] \\
&= \mathbb{E}\big[(p^k - C^k(X_t^k))Z_t^k\big] G^{k\prime}(y_{t-1}^k) - \mathbb{E}\big[R_t Z_t^k\big] G^{k\prime}(y_{t-1}^k)
\end{aligned}
$$

With $C^k(s) = \theta^k/s$ and $X_t^k = Z_t^k G^k(y_{t-1}^k)$, the shock cancels in the first expectation:

$$
\mathbb{E}\Big[\Big(p^k - \frac{\theta^k}{Z_t^k G^k(y_{t-1}^k)}\Big) Z_t^k\Big] = p^k - \frac{\theta^k}{G^k(y_{t-1}^k)} = p^k - C^k\big(G^k(y_{t-1}^k)\big)
$$

The second expectation, however, is a covariance, because $\mathbb{E}[R_t] = 0$ and $\mathbb{E}[Z_t^k] = 1$:

$$
\begin{aligned}
V_{y^k}(\vec{y}_{t-1}) &= \frac{1}{\beta}\pi_{t+1}^k(y_{t-1}^k) - \operatorname{Cov}\big(R_t,\, Z_t^k\big)\, G^{k\prime}(y_{t-1}^k) \\
\beta V_{y^k}(\vec{y}_{t-1}) &= \pi_{t+1}^k(y_{t-1}^k) - \beta \operatorname{Cov}\big(R_t,\, Z_t^k\big)\, G^{k\prime}(y_{t-1}^k)
\end{aligned}
\tag{65}
$$

In a deterministic setting, $\operatorname{Cov}(R_t, Z_t^k) = 0$ and (65) reduces to $\beta V_{y^k} = \pi_{t+1}^k$, which is Appendix E.4. Under uncertainty, the covariance is nonzero, because R_t moves with the

shocks through both λ_t^F and ρ_t^j. The planner's single scalar ω_t can set the mean of R_t to zero, but it cannot remove its correlation with the shock vector. Given the deterministic quota,

$$\sum_i Y^i(\lambda_t^F) = \sum_i Z_t^i G^i(y_{t-1}^i) - \omega_t,$$

so λ_t^F inherits the randomness of the aggregate realized stock. The certainty-equivalent quota (20),

$$\omega_t = \sum_i G^i(y_{t-1}^i) - \sum_i Y^i(\lambda^+),$$

uses only $\mathbb{E}[Z_t^i] = 1$ and the predetermined y_{t-1}^i. It centers the distribution of λ_t^F near λ^+ and the escapements near their targets $\bar{y}^i = Y^i(\lambda^+)$; it is the quota the planner can implement before the shocks are realized.

Appendix F: Bounds of the welfare loss

Throughout, $\mathcal{M} = [\lambda_L, \lambda_u]$ is an interval that contains τ^+ and every realized shadow value λ_t^F, with $\lambda_u < \underline{p} \leq \min_i p^i$. Because the shocks are bounded, $Z_t^i \in [a, b]$, the aggregate surprise S_t in (25) is bounded, and the budget identity keeps λ_t^F in a bounded set below $\min_i p^i$. I take $\mathcal{M}$ to be such a set.

Lemma 2 (Bounds for the fisherman's response)**.** *Let $Y^i(\lambda) = \theta^i/(p^i - \lambda)$ and let Assumption 1(a) hold. Define*

$$\underline{m} := \frac{\underline{\theta}}{(\overline{p} - \lambda_L)^2} > 0, \qquad \overline{m} := \frac{\overline{\theta}}{(\underline{p} - \lambda_u)^2} < \infty. \tag{66}$$

Then, for every species i and every $\lambda, \lambda' \in \mathcal{M}$,

$$\underline{m}\,|\lambda - \lambda'| \;\leq\; \left|Y^i(\lambda) - Y^i(\lambda')\right| \;\leq\; \overline{m}\,|\lambda - \lambda'|. \tag{67}$$

Proof. The response function and its derivatives are

$$Y^{i\prime}(\lambda) = \frac{d}{d\lambda}\frac{\theta^i}{p^i - \lambda} = \frac{\theta^i}{(p^i - \lambda)^2} > 0$$
$$Y^{i\prime\prime}(\lambda) = \frac{2\theta^i}{(p^i - \lambda)^3} > 0 \qquad \text{for } \lambda < p^i$$

so Y^i is strictly increasing and $Y^{i\prime}$ is itself increasing: the response is flattest at λ_L and steepest at λ_u. Hence, on $\mathcal{M}$,

$$\begin{aligned}
\frac{\theta^i}{(p^i - \lambda_L)^2} \;\leq\; Y^{i\prime}(\lambda) \;&\leq\; \frac{\theta^i}{(p^i - \lambda_u)^2} \\
\frac{\underline{\theta}}{(\overline{p} - \lambda_L)^2} \;\leq\; Y^{i\prime}(\lambda) \;&\leq\; \frac{\overline{\theta}}{(\underline{p} - \lambda_u)^2} \qquad \text{because } \underline{\theta} \leq \theta^i \leq \overline{\theta},\ \underline{p} \leq p^i \leq \overline{p} \\
\underline{m} \;\leq\; Y^{i\prime}(\lambda) \;&\leq\; \overline{m}
\end{aligned}$$

Both constants are finite and strictly positive because $\lambda_u < \underline{p}$, and neither depends on i or on N. By the mean value theorem, for any $\lambda, \lambda' \in \mathcal{M}$ there is ζ^i between them with $Y^i(\lambda) - Y^i(\lambda') = Y^{i\prime}(\zeta^i)(\lambda - \lambda')$. Since $\zeta^i \in \mathcal{M}$, its slope lies between the two bounds, which gives (67). □

Lemma 3 (Quadratic bounds around the peak)**.** *Let Ψ be twice continuously differentiable on $\mathcal{M}$, with $\Psi'(\tau^+) = 0$ and $N\underline{c} \leq -\Psi''(m\pi) \leq N\overline{c}$ for all $m\pi \in \mathcal{M}$. Then, for every $m\pi \in \mathcal{M}$,*

$$\frac{N\underline{c}}{2}\,(m\pi - \tau^+)^2 \;\leq\; \Psi(\tau^+) - \Psi(m\pi) \;\leq\; \frac{N\overline{c}}{2}\,(m\pi - \tau^+)^2$$

Proof. By the fundamental theorem of calculus, $\Psi(\tau^+) - \Psi(m\pi) = \int_{m\pi}^{\tau^+} \Psi'(s)\,ds$. Integrate by parts with $u = \Psi'(s)$ and $dv = ds$, choosing $v = s - m\pi$:

$$\begin{aligned}
\int_{m\pi}^{\tau^+} \Psi'(s)\,ds &= \Big[\Psi'(s)\,(s-m\pi)\Big]_{s=m\pi}^{s=\tau^+} - \int_{m\pi}^{\tau^+} \Psi''(s)\,(s-m\pi)\,ds \\
&= \Psi'(\tau^+)\,(\tau^+ - m\pi) - \Psi'(m\pi)\cdot 0 - \int_{m\pi}^{\tau^+} \Psi''(s)\,(s-m\pi)\,ds \\
&= \int_{m\pi}^{\tau^+} \big(-\Psi''(s)\big)\,(s-m\pi)\,ds \qquad [\Psi'(\tau^+) = 0]
\end{aligned}$$

Case $m\pi < \tau^+$.

On $[m\pi, \tau^+]$, $s - m\pi \geq 0$, so the curvature bounds give $N\underline{c}\,(s-m\pi) \leq \big(-\Psi''(s)\big)(s-m\pi) \leq N\overline{c}\,(s-m\pi)$. Integrating, with $\int_{m\pi}^{\tau^+}(s-m\pi)\,ds = \frac{1}{2}(\tau^+ - m\pi)^2$,

$$\frac{N\underline{c}}{2}\,(\tau^+ - m\pi)^2 \;\leq\; \Psi(\tau^+) - \Psi(m\pi) \;\leq\; \frac{N\overline{c}}{2}\,(\tau^+ - m\pi)^2$$

Case $m\pi > \tau^+$.

Reverse the limits of integration:

$$\int_{m\pi}^{\tau^+} \big(-\Psi''(s)\big)(s-m\pi)\,ds = \int_{\tau^+}^{m\pi} \big(-\Psi''(s)\big)(m\pi - s)\,ds,$$

where now $m\pi - s \geq 0$ on $[\tau^+, m\pi]$. The same bounds and $\int_{\tau^+}^{m\pi}(m\pi - s)\,ds = \frac{1}{2}(m\pi - \tau^+)^2$ give the result. The case $m\pi = \tau^+$ is trivial. □

Appendix G: Per-period and per-species welfare

Lemma 4 (No species-by-species maximum)**.** *At the basket escapement,* $w^{i\prime}(\bar{y}^i) = \pi^i_{t+1}(\bar{y}^i) - \tau^+$*, and its sign is the sign of* $\tau^{i*} - \tau^+$*. It is zero for every* i *if and only if all species-specific optimal taxes coincide.*

Proof. By (23) and the fisherman's condition $p^i - C^i(\bar{y}^i) = \tau^+$,

$$w^{i\prime}(\bar{y}^i) = \pi^i_{t+1}(\bar{y}^i) - \big(p^i - C^i(\bar{y}^i)\big) = \pi^i_{t+1}(\bar{y}^i) - \tau^+$$

Because Y^i is strictly increasing and $y^{i*} = Y^i(\tau^{i*})$, $\tau^+ < \tau^{i*}$ if and only if $\bar{y}^i < y^{i*}$: species i is overharvested under the basket. Then (50) gives $\pi^i_{t+1}(\bar{y}^i) > p^i - C^i(\bar{y}^i) = \tau^+$, so $w^{i\prime}(\bar{y}^i) > 0$. The case $\tau^+ > \tau^{i*}$ is symmetric. All derivatives vanish if and only if $\tau^{i*} = \tau^+$ for every i, the first-best case of Corollary 1.3. □

In a period in which the realized shadow value moves an overharvested species toward its own optimum, that species earns more under the quota than under the tax. These first-order gains and losses cancel across species: by Appendix D, $\mathcal{W}'(\tau) = -\big(\sum_i Y^{i\prime}(\tau)\big) g(\tau)$ and $g(\tau^+) = 0$, so $\sum_i w^{i\prime}(\bar{y}^i)\, Y^{i\prime}(\tau^+) = \mathcal{W}'(\tau^+) = 0$. The comparison between instruments is therefore decided at second order, by the curvature of $\mathcal{W}$ (Proposition 5).